PCA and Latent Factor Models

MFE5340 - AI in FE: Quantitative Investment

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1 Motivation from Asset Pricing

As we have seen from the previous lecture on Factor Models and Empirical Asset Pricing, the (excess) return of asset i at time t can be represent by the following linear model (I am dropping superscript e for notational simplicity):

$$R_{it} = \boldsymbol{\beta}_i^\top \boldsymbol{f}_t + e_{it}, \tag{1}$$

where $K \times 1$ vector $\boldsymbol{\beta}_i$ represent factor exposures of asset i, $K \times 1$ vector $\boldsymbol{f}_t$ represents the excess returns of those K factors at time t, and e_{it} is the zero-mean idiosyncratic error, such that $\mathbb{E}[e|\boldsymbol{f}] = 0$. Taking the covariance of both sides gives:

$$\boldsymbol{\Sigma}_r = \boldsymbol{\beta}\boldsymbol{\Sigma}_f\boldsymbol{\beta}^\top + \boldsymbol{\Sigma}_e, \tag{2}$$

where $\boldsymbol{\Sigma}_r$ and $\boldsymbol{\Sigma}_f$ are the covariance matrices of assets and factors, respectively. $\boldsymbol{\Sigma}_e$ is the covariance matrix of the error terms. Ross (1976) assumes that the random disturbances of different assets are uncorrelated, resulting in $\boldsymbol{\Sigma}_e$ being a diagonal matrix. A factor model satisfying this condition is referred to as an *exact* factor model. While factor models are elegant, their challenge lies in the identification of factors. There are two main approaches in this regard and both have gained a lot of attention.

The first approach assumes either factors or factor exposures are **known** and **observable**. The academia and the industry take different stances in this regard. Specifically, the academia considers that factors are tradable portfolios that are constructed by using some firm characteristics, such as the HML and SMB factors of Fama and French (1993). In this case, factor realizations are the excess returns of those tradable portfolios, and factor exposures are determined by regressing assets' excess returns on factors.

The second approach, pioneered by Chamberlain and Rothschild (1983) and Connor and Korajczyk (1986), assumes both factors and factor exposures are **latent**. Its starting point is the covariance matrix equation (2). This equation implies that the covariance of the factors should be able to explain a large portion of the assets' covariance. Therefore, we can also estimate the latent factors and their corresponding exposures from the assets' covariance matrix. When it comes to analyzing the covariance matrix, the most straightforward method is factor analysis. Roll and Ross (1980) employed this method to analyze early U.S. stock return data and discovered three to four factors that successfully explained the expected returns of assets to a certain extent.

Subsequently, Chamberlain and Rothschild (1983) relaxed Ross's assumption about $\boldsymbol{\Sigma}_e$, allowing for weak correlations between random disturbances, leading to an *approximate* factor model. For this model, Chamberlain and Rothschild (1983) pointed out that using Principal Component Analysis (PCA) instead of factor analysis could yield similar results. Specifically, they suggested that extracting the eigenvectors of the covariance matrix is as good a method as any for finding approximate factor structures. In subsequent research, Connor and Korajczyk (1986, 1988) proposed

asymptotic principal components analysis (APCA) when the number of assets in the cross-section increases. These studies established the foundation for using PCA to study latent factors.

We can understand Chamberlain and Rothschild (1983)'s suggestion intuitively as follows. In the context of asset pricing, it is reasonable to assume that asset returns are determined by a few latent factors, leading to a covariance matrix of asset returns where only a few eigenvalues are significantly large. In this scenario, the eigenvectors corresponding to these large eigenvalues can serve as approximations for our factor loadings, effectively providing a good approximation of the unknown latent factors. Given this premise, PCA, which is computationally and conceptually simpler than factor analysis, becomes an important tool for finding an approximate factor structure.

Using PCA to analyze asset pricing has several benefits. PCA is a powerful technique that helps in reducing the dimensionality of the data while preserving as much variance as possible. This is particularly useful in asset pricing where financial data often involves a large number of correlated variables. By identifying and retaining the principal components that capture the most variance, PCA helps in simplifying the data structure, making it easier to identify underlying risk factors that drive asset returns. Another benefit is that PCA does not require any specific distributional assumptions about the data. **Unlike factor analysis, which often assumes normality, PCA is purely data-driven.** This flexibility makes PCA a versatile tool that can be applied to various types of financial data without needing to satisfy stringent statistical assumptions.

2 Principal Component Analysis

We have explained the motivation for discovering latent factors via PCA. In this section, we delve into the technical details of PCA.

2.1 Intuitive Explanation

PCA is a powerful statistical technique used to uncover the underlying structure in a dataset by transforming it into a new coordinate system. This transformation highlights the directions (principal components) along which the data varies the most.

Before applying PCA, it is crucial to **center the data** by subtracting the mean of each variable (i.e., demean). This step ensures that the data is considered relative to its center, which is the mean of the dataset. Centering the data removes any biases due to the mean and allows PCA to focus on the variations and relationships between the variables. By centering, we are effectively shifting the origin of our coordinate system to the mean of the data, ensuring that the principal components reflect the true directions of maximum variance.

Intuitively, **PCA can be viewed as a rotational transformation of the data**. It identifies the directions in which the data points exhibit the largest variance from the center. These directions are the principal components. By projecting the data onto these principal components, PCA transforms the dataset into a new coordinate system where the axes correspond to the directions of

maximum variability. This transformation simplifies the complexity of the data, making it easier to identify patterns and relationships. The principal components are ordered by the amount of variance they capture, with the first principal component capturing the most variance, followed by the second, and so on.

Figure 1 illustrates the importance of centering the data before applying PCA. The plot on the left shows the **incorrect** PCA result without demeaning the data, while the plot on the right shows the **correct** PCA result with the data properly demeaned.

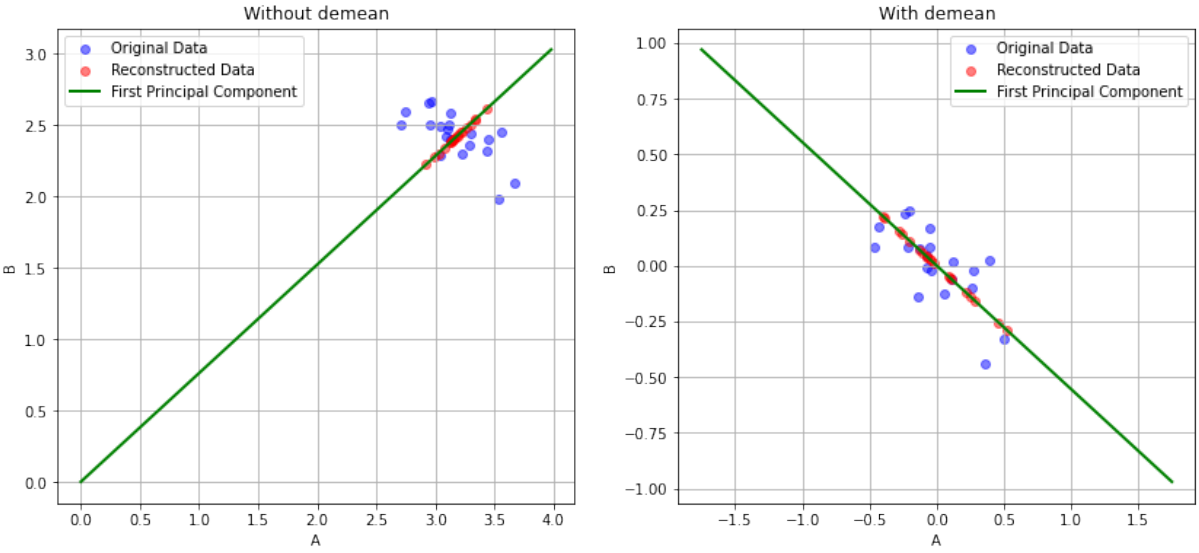


Figure 1: PCA Non-demean vs. Demean

- **Left Plot (Without Demeaning)**: The original data is shown in blue, the reconstructed data is in red, and the first principal component is the green line. Without demeaning, the variation of the points is calculated relative to the origin of the coordinate axes, rather than the center of the data points themselves. Because PCA is a rotational transformation that must pass through the origin, it yields incorrect results when the data is not centered. The principal component shown does not accurately capture the direction of maximum variance, failing to reflect the true structure of the data.

- **Right Plot (With Demeaning)**: Again, the original data is shown in blue, the reconstructed data is in red, and the first principal component is the green line. With demeaning, the data has been correctly centered, so the new origin reflects the center of the data points. The variation is now correctly calculated relative to this center. After the PCA transformation, the new coordinate axes also pass through the center of the data (the new origin after centering), which allows the principal component to correctly capture the direction of maximum variance, reflecting the true underlying structure of the data.

2.2 Mathematical Derivation

Assume we have N assets and let the $N \times 1$ random vector $\boldsymbol{r} = [r_1, \cdots, r_N]^\top$ represent the returns of these assets for a given period. Further, let the $N \times N$ matrix $\boldsymbol{\Sigma}_r$ be the covariance matrix of these assets. It is important to note that the covariance matrix $\boldsymbol{\Sigma}_r$ represents the variation of the data relative to its center (mean), emphasizing that PCA is concerned with the variation around the mean rather than absolute values.

PCA proceeds as follows: We aim to find a linear combination of these N assets, i.e., $f_1 = \boldsymbol{w}_1^\top \boldsymbol{r}$, such that the variance of f_1 (i.e., $\text{var}(f_1)$) is maximized. The constructed f_1 has a clear intuitive explanation: it can be viewed as a portfolio of the original N assets weighted by $\boldsymbol{w}_1$, and this portfolio is viewed as a factor. Therefore, we seek to find the weight $\boldsymbol{w}_1$ such that factor f_1 can explain as much variance of the original data as possible. Here, the variance being maximized is the variation around the center of the data, which is crucial for capturing the underlying structure. Of course, when trying to maximize it, we can always change $\text{var}(f_1)$ by scaling up $\boldsymbol{w}_1$. Hence, we need to impose the following constraint when solving for $\boldsymbol{w}_1$: $\boldsymbol{w}_1^\top \boldsymbol{w}_1 = 1$.

According to the definition of variance, $\text{var}(f_1)$ can be written as $\boldsymbol{w}_1' \boldsymbol{\Sigma}_r \boldsymbol{w}_1$. Therefore, the optimization problem for solving $\boldsymbol{w}_1$ is as follows:

$$\begin{aligned} \max \quad & \boldsymbol{w}_1^\top \boldsymbol{\Sigma}_r \boldsymbol{w}_1, \\ \text{subject to} \quad & \boldsymbol{w}_1^\top \boldsymbol{w}_1 = 1. \end{aligned} \tag{3}$$

The Lagrange function ($\mathcal{L}$) for this optimization problem is:

$$\mathcal{L} = \boldsymbol{w}_1^\top \boldsymbol{\Sigma}_r \boldsymbol{w}_1 - \lambda_1 (\boldsymbol{w}_1^\top \boldsymbol{w}_1 - 1). \tag{4}$$

Its first-order condition (with respect to $\boldsymbol{w}_1$) is:

$$\boldsymbol{\Sigma}_r \boldsymbol{w}_1 = \lambda_1 \boldsymbol{w}_1. \tag{5}$$

From the above result, we can see that $\boldsymbol{w}_1$ is an eigenvector of $\boldsymbol{\Sigma}_r$, and λ_1 is the corresponding eigenvalue. Since we want to maximize $\text{var}(f_1)$, $\boldsymbol{w}_1$ corresponds to the eigenvector of $\boldsymbol{\Sigma}_r$ with the largest eigenvalue. With $\boldsymbol{w}_1$ found, the first principal component of a data vector $\boldsymbol{r}$ is simply the factor $f_1 = \boldsymbol{w}_1^\top \boldsymbol{r}$. Intuitively, this can be understood as mapping the original N-dimensional vector to the 1-dimensional first principal component f_1, which is the portfolio of these assets with the largest variance.

Once we have the first principal component, we can similarly find the next set of weights $\boldsymbol{w}_2$ and thereby determine the second principal component. The selection criterion for $\boldsymbol{w}_2$ is to maximize $\boldsymbol{w}_2^\top \boldsymbol{\Sigma}_r \boldsymbol{w}_2$ subject to being orthogonal to $\boldsymbol{w}_1$ and $\boldsymbol{w}_2^\top \boldsymbol{w}_2 = 1$. It can be proven that $\boldsymbol{w}_2$ corresponds to the eigenvector of $\boldsymbol{\Sigma}_r$ associated with the second largest eigenvalue.

Finally, by extending the above derivation, we can obtain all the principal components. Therefore, the PCA concept can be summarized as (Tsay 2010):

1. The first principal component of $\boldsymbol{r}$ is the linear combination (factor) $f_1 = \boldsymbol{w}_1^\top \boldsymbol{r}$ that maximizes $\text{var}(f_1)$ subject to $\boldsymbol{w}_1^\top \boldsymbol{w}_1 = 1$.

2. The second principal component of $\boldsymbol{r}$ is the linear combination (factor) $f_2 = \boldsymbol{w}_2^\top \boldsymbol{r}$ that maximizes $\text{var}(f_2)$ subject to $\boldsymbol{w}_2^\top \boldsymbol{w}_2 = 1$ and being orthogonal to $\boldsymbol{w}_1$.

3. The k-th principal component of $\boldsymbol{r}$ is the linear combination (factor) $f_k = \boldsymbol{w}_k^\top \boldsymbol{r}$ that maximizes $\text{var}(f_k)$ subject to $\boldsymbol{w}_k^\top \boldsymbol{w}_k = 1$ and being orthogonal to $\boldsymbol{w}_j$ for $j = 1, \cdots, k-1$.

Factor Exposure in Asset Pricing

Given that these factors are obtained through PCA, they are often referred to as principal component (PC) factors in academic literature (e.g., Nagel 2021). What are the factor exposures (or factor loadings) of these N assets to these factors? Intuitively, for factor k, the exposure of the assets $\boldsymbol{\beta}_k = [\beta_{1k}, \beta_{2k}, \cdots, \beta_{Nk}]^\top$ should be equal to $\boldsymbol{w}_k = [w_{1k}, w_{2k}, \cdots, w_{Nk}]^\top$. This is indeed the case:

$$\beta_{ik} = \frac{\text{cov}(r_i, f_k)}{\text{var}(f_k)} = \frac{\text{cov}(r_i, \sum_j w_{jk} r_j)}{\lambda_k} = \frac{\sum_j w_{jk} \text{cov}(r_i, r_j)}{\lambda_k}. \tag{6}$$

According to the definition of eigenvectors:

$$\boldsymbol{\Sigma}_r \boldsymbol{w}_k = \lambda_k \boldsymbol{w}_k \Rightarrow \text{for each row } i, \sum_j w_{jk} \text{cov}(r_i, r_j) = \lambda_k w_{ik}. \tag{7}$$

Therefore

$$\beta_{ik} = \frac{\lambda_k w_{ik}}{\lambda_k} = w_{ik}, \quad i = 1, \cdots, N. \tag{8}$$

Lack of First Moment Information

A factor return series $\boldsymbol{f}$ with a non-zero mean implies that we can use the average returns of factors (i.e., the estimated factor risk premia) for asset pricing. However, from the perspective of PCA's motivation, when constructing these factors, we only utilized the second **central** moment information of the original data relative to its mean (i.e., variance), without explicitly using the first moment information. As we will see in Section 6, this deficiency provides an opportunity for subsequent modifications to the PCA-based latent factor models.

2.3 Examples

The following example finds the first PC for a two-dimensional data. In this example, I used the method of eigen-decomposition on the covariance matrix to solve for the eigenvectors. Sometimes, we can also directly use `sklearn`'s PCA class to perform PCA. However, it is helpful to mention the following extracted from `sklearn`'s documentation:

Linear dimensionality reduction using Singular Value Decomposition (SVD) of the data to project it to a lower dimensional space. The input data is centered but not scaled for each feature before applying the SVD.

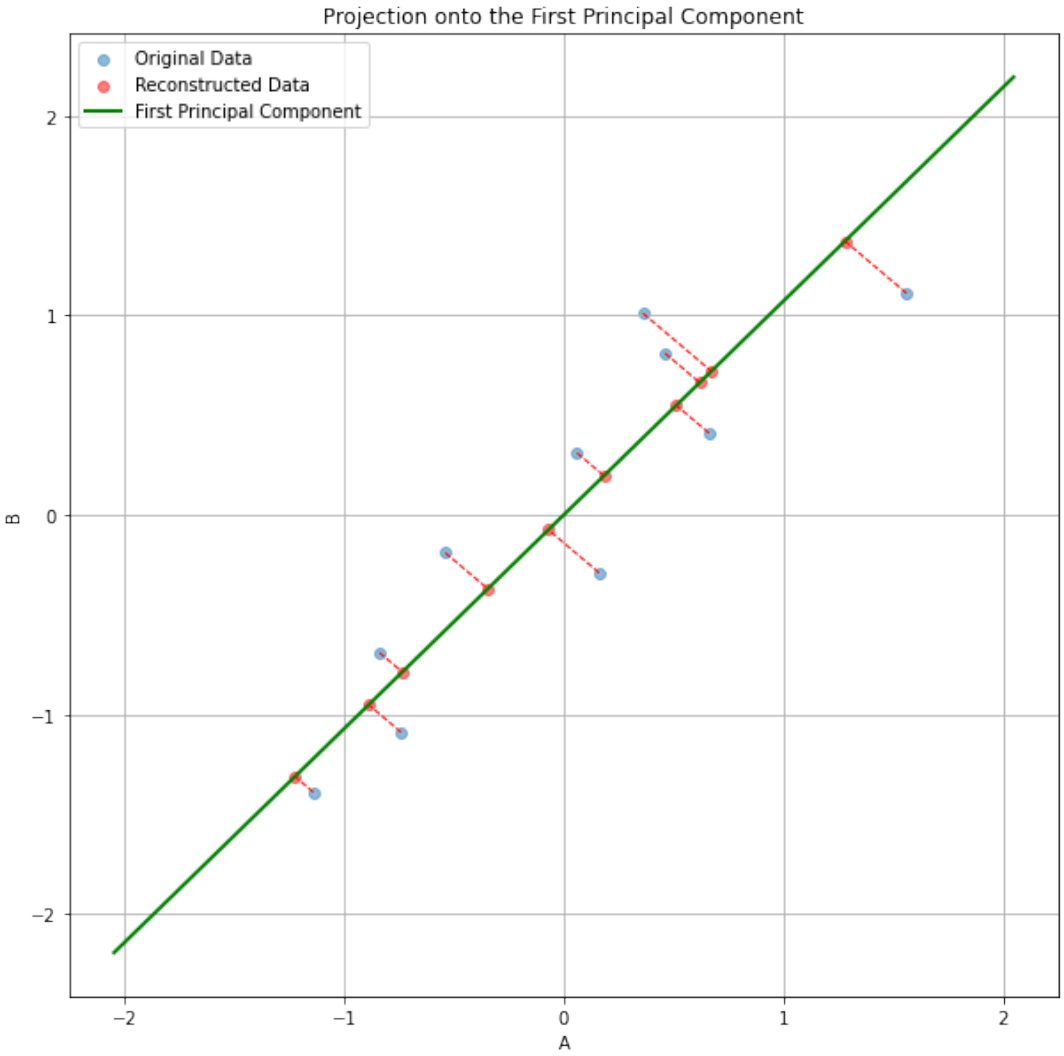


Figure 2: PCA Example

This means that instead of applying eigen-decomposition, it performs SVD on the demeaned original data (and it centers the data for you in case you forget about it! This is because SVD results on demeaned and non-demeaned data are different!). We will explain the equivalence between the two approaches and further leverage this property in Section 5. In this example, I also compare the demeaned factors $\tilde{\mathbf{F}}$ and the non-demeaned version $\mathbf{F}$ (which is derived by using `np.dot(R, eig_vecs)`).

2.4 Best Low-Rank Approximation

It should be clear to us that PCA aims to explain as much variation of the original data (i.e., the structure of $\mathbf{\Sigma}_r$) as possible using a few principal components (i.e., a few factors). Therefore, even though we can obtain a total of N principal components for the original data $\tilde{\mathbf{R}}$ with full rank columns (I am using $T \times N$ matrix $\tilde{\mathbf{R}}$ to represent the mean-centered data), in practice, we often select only the first K components because they already explain the majority of the variation. In this sense, PCA achieves **dimensionality reduction**, and we are effectively using K principal

components to perform a **low-rank approximation** of the original data:

$$\tilde{\mathbf{R}} = \tilde{\mathbf{F}}_K \mathbf{\Lambda}_K^\top + \mathbf{E}, \tag{9}$$

where $\mathbf{\Lambda}_K$ represents an $N \times K$ matrix composed of the first K eigenvectors, $\tilde{\mathbf{F}}_K = \tilde{\mathbf{R}}\mathbf{\Lambda}_K$ is the corresponding $T \times K$ matrix of factor returns, and $\mathbf{E}$ represents the error term.

Temporarily setting aside PCA, we can view $\tilde{\mathbf{R}} = \tilde{\mathbf{F}}_K \mathbf{\Lambda}_K^\top + \mathbf{E}$ as a linear regression model. Therefore, for any given $T \times K$ design matrix $\mathbf{\Lambda}_K$, we can solve for the optimal $\tilde{\mathbf{F}}_K = \tilde{\mathbf{R}}\mathbf{\Lambda}_K(\mathbf{\Lambda}_K^\top \mathbf{\Lambda}_K)^{-1}$ by minimizing $\|\mathbf{E}\|_F^2$ (where $\|.\|_F$ is the Frobenius norm). When $\mathbf{\Lambda}_K$ consists of the first K eigenvectors, due to the orthogonality of the eigenvectors, $\mathbf{\Lambda}_K^\top \mathbf{\Lambda}_K = \mathbf{I}_K$, so $\tilde{\mathbf{F}}_K = \tilde{\mathbf{R}}\mathbf{\Lambda}_K(\mathbf{\Lambda}_K^\top \mathbf{\Lambda}_K)^{-1} = \tilde{\mathbf{R}}\mathbf{\Lambda}_K$.

It can be proven that among all design matrices, when the columns of $\mathbf{\Lambda}_K$ are the first K eigenvectors, the resulting low-rank approximation $\tilde{\mathbf{F}}_K \mathbf{\Lambda}_K$ is the best low-rank approximation. The proof starts with SVD of the demeaned matrix $\tilde{\mathbf{R}}$:

$$\tilde{\mathbf{R}} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top, \tag{10}$$

where $\mathbf{U}$ is a $T \times T$ orthogonal matrix, $\mathbf{\Sigma}$ is a $T \times N$ diagonal matrix with singular values, and $\mathbf{V}$ is an $N \times N$ orthogonal matrix. We can partition $\mathbf{U}$, $\mathbf{\Sigma}$, and $\mathbf{V}$ as follows: $\mathbf{U} = [\mathbf{U}_K, \mathbf{U}_{\text{rest}}]$, $\mathbf{V} = [\mathbf{V}_K, \mathbf{V}_{\text{rest}}]$, and

$$\mathbf{\Sigma} = \begin{bmatrix} \mathbf{\Sigma}_K & \mathbf{0} \\ \mathbf{0} & \mathbf{\Sigma}_{\text{rest}} \end{bmatrix}. \tag{11}$$

where $\mathbf{U}_K$ is $T \times K$ containing the first K columns of $\mathbf{U}$, $\mathbf{V}_K$ is $N \times K$ containing the first K columns of $\mathbf{V}$, and $\mathbf{\Sigma}_K$ is $K \times K$ containing the largest K singular values. It is known that the truncated SVD, i.e., $\mathbf{U}_K \mathbf{\Sigma}_K \mathbf{V}_K^\top$, provides the best low-rank approximation in the Frobenius norm, and we shall link this to PCA.

Since $\tilde{\mathbf{R}}$ is demeaned, its covariance matrix is propotional to $\tilde{\mathbf{R}}^\top \tilde{\mathbf{R}}$. By using the SVD of $\tilde{\mathbf{R}}$, this is equal to $\mathbf{V}(\mathbf{\Sigma}^\top \mathbf{\Sigma})\mathbf{V}^\top$. Hence, the orthogonal matrix $\mathbf{V}$ is just the eigenvectors of the covariance matrix, and therefore $\mathbf{\Lambda}_K = \mathbf{V}_K$, and $\mathbf{F}_K = \tilde{\mathbf{R}}\mathbf{V}_K$. Given the equivalence between SVD on $\tilde{\mathbf{R}}$ and eigen-decomposition on the covariance matrix of $\tilde{\mathbf{R}}$, `sklearn`'s PCA implementation performs SVD on the demeaned original data rather than directly computing the eigenvalues and eigenvectors of the covariance matrix.

Now, as long as we can show that $\tilde{\mathbf{R}}\mathbf{V}_K = \mathbf{U}_K \mathbf{\Sigma}_K$, we are done. Substituting $\tilde{\mathbf{R}} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ into the expression for $\mathbf{F}_K$:

$$\tilde{\mathbf{R}}\mathbf{V}_K = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top \mathbf{V}_K. \tag{12}$$

Given the partitioned form of $\mathbf{V}$, we have:

$$\mathbf{V}^\top \mathbf{V}_K = \begin{bmatrix} \mathbf{V}_K^\top \\ \mathbf{V}_{\text{rest}}^\top \end{bmatrix} \mathbf{V}_K = \begin{bmatrix} \mathbf{I}_K \\ \mathbf{0} \end{bmatrix}. \tag{13}$$

Thus:

$$\tilde{\mathbf{R}}\mathbf{V}_K = \mathbf{U}\boldsymbol{\Sigma}\begin{bmatrix}\mathbf{I}_K\\ \mathbf{0}\end{bmatrix} = \mathbf{U}\begin{bmatrix}\boldsymbol{\Sigma}_K\\ \mathbf{0}\end{bmatrix} \tag{14}$$

Finally, since:

$$\mathbf{U}\begin{bmatrix}\boldsymbol{\Sigma}_K\\ \mathbf{0}\end{bmatrix} = [\mathbf{U}_K, \mathbf{U}_{\text{rest}}]\begin{bmatrix}\boldsymbol{\Sigma}_K\\ \mathbf{0}\end{bmatrix} = \mathbf{U}_K\boldsymbol{\Sigma}_K + \mathbf{U}_{\text{rest}}\mathbf{0} = \mathbf{U}_K\boldsymbol{\Sigma}_K, \tag{15}$$

we have:

$$\mathbf{F} = \tilde{\mathbf{R}}\mathbf{V}_K = \mathbf{U}_K\boldsymbol{\Sigma}_K. \tag{16}$$

Hence, choosing the first K eigenvectors of $\tilde{\mathbf{R}}^\top\tilde{\mathbf{R}}$ as the design matrix ensures $\|\mathbf{E}\|_F^2$ is minimized.

3 PCA and Alternating Least Squares

In this section, we continue our discussion in Section 2.4, but from a different angle. Section 2.4 says, performing PCA is equivalent to minimizing the error, i.e., $\|\mathbf{E}\|_F^2$, of the following linear regression model:

$$\tilde{\mathbf{R}} = \tilde{\mathbf{F}}_K\boldsymbol{\Lambda}_K^\top + \mathbf{E}. \tag{17}$$

From the above perspective, we can view PCA as solving a linear regression problem using the least squares method. For instance, using the first K eigenvectors as the design matrix, we can compute $\tilde{\mathbf{F}}_K$ via $\tilde{\mathbf{F}}_K = \tilde{\mathbf{R}}\boldsymbol{\Lambda}_K(\boldsymbol{\Lambda}_K^\top\boldsymbol{\Lambda}_K)^{-1}$.

Now, let's consider an alternative approach: can we obtain the same solution as direct PCA by solving a linear regression without directly computing the eigenvectors? The answer is yes, and this method is known as Alternating Least Squares (ALS). ALS is an **iterative** optimization technique commonly used for solving least squares problems, particularly in the context of matrix factorization. In the PCA context, we can leverage ALS to iteratively update the factor loadings $\boldsymbol{\Lambda}_K$ and the factor returns $\tilde{\mathbf{F}}_K$ until convergence is achieved.

Therefore, in ALS, we assume both $\boldsymbol{\Lambda}_K$ and $\tilde{\mathbf{F}}_K$ are unknown and initialize them randomly. Then, with the goal of minimizing $\|\mathbf{E}\|_F^2$, we alternately update $\boldsymbol{\Lambda}_K^{LS}$ and $\tilde{\mathbf{F}}_K^{LS}$ (superscript LS indicates results derived from ALS): given $\boldsymbol{\Lambda}_K^{LS}$, we update $\tilde{\mathbf{F}}_K^{LS}$ for the next iteration; given $\tilde{\mathbf{F}}_K^{LS}$, we update $\boldsymbol{\Lambda}_K^{LS}$ for the next iteration, until both sequences converge. Section 2.4 has already demonstrated that PCA provides the optimal low-rank approximation for the given original data $\tilde{\mathbf{R}}$. Therefore, if the ALS iterations can also lead to the global optimal solution, the $\boldsymbol{\Lambda}_K^{LS}$ and $\tilde{\mathbf{F}}_K^{LS}$ obtained by this method should be consistent with those obtained by directly solving PCA.

3.1 Alternating Least Squares

Now let's look at the specific steps of ALS:

1. Initialization: Begin with an initial guess for the factor loadings $\boldsymbol{\Lambda}_K^{LS}$. For example, this can be a random initialization.

2. Fix $\mathbf{\Lambda}_K^{LS}$ and solve for $\tilde{\mathbf{F}}_K^{LS}$: With $\mathbf{\Lambda}_K^{LS}$ fixed, solve the least squares problem for $\tilde{\mathbf{F}}_K^{LS}$. This is achieved by minimizing $\|\mathbf{E}\|_F^2$:

$$\tilde{\mathbf{F}}_K^{LS} = \tilde{\mathbf{R}}\mathbf{\Lambda}_K^{LS}\left[(\mathbf{\Lambda}_K^{LS})^\top(\mathbf{\Lambda}_K^{LS})\right]^{-1}. \tag{18}$$

   Here, $\tilde{\mathbf{F}}_K^{LS}$ is updated by projecting the demeaned returns $\tilde{\mathbf{R}}$ onto the fixed factor loadings $\mathbf{\Lambda}_K^{LS}$.

3. Fix $\tilde{\mathbf{F}}_K^{LS}$ and solve for $\mathbf{\Lambda}_K^{LS}$: With $\tilde{\mathbf{F}}_K^{LS}$ fixed, update the factor loadings $\mathbf{\Lambda}_K^{LS}$ by solving the least squares problem:

$$\mathbf{\Lambda}_K^{LS} = \tilde{\mathbf{R}}^\top\tilde{\mathbf{F}}_K^{LS}\left[(\tilde{\mathbf{F}}_K^{LS})^\top(\tilde{\mathbf{F}}_K^{LS})\right]^{-1}. \tag{19}$$

   This step involves projecting the transposed demeaned returns, $\tilde{\mathbf{R}}^\top$, onto the fixed factor returns $\tilde{\mathbf{F}}_K^{LS}$.

4. Iterate: Repeat steps 2 and 3 iteratively until convergence. Convergence is typically determined by checking if the change in the error $\|\mathbf{E}\|_F^2$ or factor loadings $\mathbf{\Lambda}_K^{LS}$ as well as the factor matrix $\tilde{\mathbf{F}}_K^{LS}$ fall below a specified tolerance level.

3.2 Orthogonalization and Rotation

When using ALS to solve PCA, there are two important issues to note.

First, when updating $\mathbf{\Lambda}_K^{LS}$ given $\tilde{\mathbf{F}}_K^{LS}$, its formula does not guarantee that the resulting $\mathbf{\Lambda}_K^{LS}$ will be orthogonal. To address this, an orthogonalization step can be added after each iteration of $\mathbf{\Lambda}_K^{LS}$. This step transforms the columns of $\mathbf{\Lambda}_K^{LS}$ into a set of orthogonal bases that span the column space of $\mathbf{\Lambda}_K^{LS}$. This can resolve the non-orthogonality issue of $\mathbf{\Lambda}_K^{LS}$ in each iteration.

Second, when updating $\tilde{\mathbf{F}}_K^{LS}$ given $\mathbf{\Lambda}_K^{LS}$ (after orthogonalization), its formula does not ensure that the diagonal elements of the covariance matrix of $\tilde{\mathbf{F}}_K^{LS}$ are ordered from largest to smallest (or even that the covariance matrix of $\tilde{\mathbf{F}}_K^{LS}$ is diagonal). This is because $(\tilde{\mathbf{F}}_K^{LS})(\mathbf{\Lambda}_K^{LS})^\top$ possesses **rotational invariance**, meaning they are identifiable only up to a linear rotation. To see why, consider that for any orthogonal matrix $\mathbf{Q}$, the following holds:

$$(\tilde{\mathbf{F}}_K^{LS}\mathbf{Q})(\mathbf{\Lambda}_K^{LS}\mathbf{Q})^\top = \tilde{\mathbf{F}}_K^{LS}\mathbf{Q}\mathbf{Q}^\top(\mathbf{\Lambda}_K^{LS})^\top = (\tilde{\mathbf{F}}_K^{LS})(\mathbf{\Lambda}_K^{LS})^\top. \tag{20}$$

Therefore, even though we enforce orthogonality on $\mathbf{\Lambda}_K^{LS}$ during the iterations, without additional constraints on $\tilde{\mathbf{F}}_K^{LS}$, $\mathbf{\Lambda}_K^{LS}$ may not converge to $\mathbf{\Lambda}_K$ and $\tilde{\mathbf{F}}_K^{LS}$ may not converge to $\tilde{\mathbf{F}}_K$, but only to a rotated version of $\tilde{\mathbf{F}}_K$.

Given the above discussion, we need to rotate $\tilde{\mathbf{F}}_K^{LS}$ after each iteration to ensure that the diagonal elements of its covariance matrix are ordered from largest to smallest. This can be achieved by performing eigen-decomposition on the covariance matrix of $\tilde{\mathbf{F}}_K^{LS}$. Based on the discussion, we incorporate an orthogonalization step in the calculation of $\mathbf{\Lambda}_K^{LS}$ and a rotation step in the calculation

of $\tilde{\mathbf{F}}_K^{LS}$ during the alternating iterations.

3.3 Discussion

When using ALS for solving PCA, each local optimization problem, given $\tilde{\mathbf{F}}_K^{LS}$ or $\mathbf{\Lambda}_K^{LS}$, is indeed a convex optimization problem. This convexity ensures that the local solutions found at each step are optimal for the subproblems. However, when these subproblems are considered together in the context of ALS, the overall problem is not necessarily convex. The alternating nature of the updates means that while each step is locally optimal, there is no guarantee that the sequence of updates will converge to the global optimum of the original PCA problem.

Guaranteeing convergence to the global optimum using ALS is challenging due to the non-convex nature of the overall problem. However, several strategies can be employed to improve the chances of achieving a good solution:

1. Initialization: Using a good initialization strategy, such as starting with the results of a preliminary PCA, can significantly enhance convergence quality and speed.

2. Regularization: Adding regularization terms to the optimization problems can help stabilize the solutions and avoid overfitting, leading to better convergence properties.

3. Multiple Runs: Running the ALS algorithm multiple times with different initializations and selecting the best solution based on the objective function value can also improve the chances of finding a near-optimal solution.

While direct PCA has a closed-form solution through eigen-decomposition, there are scenarios where ALS becomes more attractive or necessary. Recent developments in PCA methodology have introduced modifications that do not have analytical solutions and therefore require iterative approaches like ALS. One notable example is the instrumented PCA (IPCA) method proposed by Kelly, Pruitt, and Su (2019). IPCA and other advanced variations improve upon traditional PCA by incorporating additional information into the factor model. These methods often lead to optimization problems that are not solvable by direct eigen-decomposition, necessitating iterative techniques like ALS. We will talk more about IPCA in Section 6. According to Kelly, Pruitt, and Su (2019), convergence to the global optimum is not a concern in their approach.

4 Sample Covariance Matrix

Let the $T \times N$ matrix $\mathbf{R}$ represent asset returns, where T is number of periods and N is the number of assets. Besides, let $N \times 1$ vector $\bar{\boldsymbol{R}}$ represent the sample mean returns of N assets; and $\boldsymbol{R}_t$ represent the **transpose** of the t-th row of $\mathbf{R}$ (so $\boldsymbol{R}_t$ itself is a $N \times 1$ column vector!). Starting with

the definition, the (biased) sample covariance $\mathbf{S}$ is defined as:

$$\begin{aligned}
\mathbf{S} &= \begin{bmatrix} \frac{1}{T}\sum_{t=1}^{T}(R_{1t}-\bar{R}_1)(R_{1t}-\bar{R}_1) & \frac{1}{T}\sum_{t=1}^{T}(R_{1t}-\bar{R}_1)(R_{2t}-\bar{R}_2) & \cdots & \frac{1}{T}\sum_{t=1}^{T}(R_{1t}-\bar{R}_1)(R_{Nt}-\bar{R}_N) \\ \frac{1}{T}\sum_{t=1}^{T}(R_{2t}-\bar{R}_2)(R_{1t}-\bar{R}_1) & \frac{1}{T}\sum_{t=1}^{T}(R_{2t}-\bar{R}_2)(R_{2t}-\bar{R}_2) & \cdots & \frac{1}{T}\sum_{t=1}^{T}(R_{2t}-\bar{R}_2)(R_{Nt}-\bar{R}_N) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{1}{T}\sum_{t=1}^{T}(R_{Nt}-\bar{R}_N)(R_{1t}-\bar{R}_1) & \frac{1}{T}\sum_{t=1}^{T}(R_{Nt}-\bar{R}_N)(R_{2t}-\bar{R}_2) & \cdots & \frac{1}{T}\sum_{t=1}^{T}(R_{Nt}-\bar{R}_N)(R_{Nt}-\bar{R}_N) \end{bmatrix} \\
&= \frac{1}{T}\begin{bmatrix} \sum_{t=1}^{T}(R_{1t}-\bar{R}_1)(R_{1t}-\bar{R}_1) & \sum_{t=1}^{T}(R_{1t}-\bar{R}_1)(R_{2t}-\bar{R}_2) & \cdots & \sum_{t=1}^{T}(R_{1t}-\bar{R}_1)(R_{Nt}-\bar{R}_N) \\ \sum_{t=1}^{T}(R_{2t}-\bar{R}_2)(R_{1t}-\bar{R}_1) & \sum_{t=1}^{T}(R_{2t}-\bar{R}_2)(R_{2t}-\bar{R}_2) & \cdots & \sum_{t=1}^{T}(R_{2t}-\bar{R}_2)(R_{Nt}-\bar{R}_N) \\ \vdots & \vdots & \ddots & \vdots \\ \sum_{t=1}^{T}(R_{Nt}-\bar{R}_N)(R_{1t}-\bar{R}_1) & \sum_{t=1}^{T}(R_{Nt}-\bar{R}_N)(R_{2t}-\bar{R}_2) & \cdots & \sum_{t=1}^{T}(R_{Nt}-\bar{R}_N)(R_{Nt}-\bar{R}_N) \end{bmatrix} \\
&= \frac{1}{T}\sum_{t=1}^{T}\begin{bmatrix} (R_{1t}-\bar{R}_1)(R_{1t}-\bar{R}_1) & (R_{1t}-\bar{R}_1)(R_{2t}-\bar{R}_2) & \cdots & (R_{1t}-\bar{R}_1)(R_{Nt}-\bar{R}_N) \\ (R_{2t}-\bar{R}_2)(R_{1t}-\bar{R}_1) & (R_{2t}-\bar{R}_2)(R_{2t}-\bar{R}_2) & \cdots & (R_{2t}-\bar{R}_2)(R_{Nt}-\bar{R}_N) \\ \vdots & \vdots & \ddots & \vdots \\ (R_{Nt}-\bar{R}_N)(R_{1t}-\bar{R}_1) & (R_{Nt}-\bar{R}_N)(R_{2t}-\bar{R}_2) & \cdots & (R_{Nt}-\bar{R}_N)(R_{Nt}-\bar{R}_N) \end{bmatrix}
\end{aligned} \tag{21}$$

4.1 First Expression

We can stack $R_{it}-\bar{R}_i$ for all N assets together for a given t:

$$\boldsymbol{R}_t - \bar{\boldsymbol{R}} = \begin{bmatrix} R_{1t}-\bar{R}_1 \\ R_{2t}-\bar{R}_2 \\ \vdots \\ R_{Nt}-\bar{R}_N \end{bmatrix}, \tag{22}$$

and therefore:

$$(\boldsymbol{R}_t - \bar{\boldsymbol{R}})(\boldsymbol{R}_t - \bar{\boldsymbol{R}})^\top = \begin{bmatrix} (R_{1t}-\bar{R}_1)(R_{1t}-\bar{R}_1) & (R_{1t}-\bar{R}_1)(R_{2t}-\bar{R}_2) & \cdots & (R_{1t}-\bar{R}_1)(R_{Nt}-\bar{R}_N) \\ (R_{2t}-\bar{R}_2)(R_{1t}-\bar{R}_1) & (R_{2t}-\bar{R}_2)(R_{2t}-\bar{R}_2) & \cdots & (R_{2t}-\bar{R}_2)(R_{Nt}-\bar{R}_N) \\ \vdots & \vdots & \ddots & \vdots \\ (R_{Nt}-\bar{R}_N)(R_{1t}-\bar{R}_1) & (R_{Nt}-\bar{R}_N)(R_{2t}-\bar{R}_2) & \cdots & (R_{Nt}-\bar{R}_N)(R_{Nt}-\bar{R}_N) \end{bmatrix}. \tag{23}$$

Hence, the first expression for $\mathbf{S}$ is:

$$\mathbf{S} = \frac{1}{T}\sum_{t=1}^{T}(\boldsymbol{R}_t - \bar{\boldsymbol{R}})(\boldsymbol{R}_t - \bar{\boldsymbol{R}})^\top. \tag{24}$$

4.2 Second Expression

We can write $T \times N$ matrix $\mathbf{R} - \mathbf{1}_T \bar{\boldsymbol{R}}^\top$ as follows:

$$\underbrace{\mathbf{R}}_{T\times N} - \underbrace{\mathbf{1}_T}_{T\times 1}\underbrace{\bar{\boldsymbol{R}}^\top}_{1\times N} = \begin{bmatrix} R_{11}-\bar{R}_1 & R_{21}-\bar{R}_2 & \cdots & R_{N1}-\bar{R}_N \\ R_{12}-\bar{R}_1 & R_{22}-\bar{R}_2 & \cdots & R_{N2}-\bar{R}_N \\ \vdots & \vdots & \ddots & \vdots \\ R_{1T}-\bar{R}_1 & R_{2T}-\bar{R}_2 & \cdots & R_{NT}-\bar{R}_N \end{bmatrix}, \tag{25}$$

where $\mathbf{1}_T$ is a $T \times 1$ vector of ones. The transpose of $\boldsymbol{R} - \mathbf{1}_T \bar{\boldsymbol{R}}^\top$ is:

$$(\mathbf{R} - \mathbf{1}_T \bar{\mathbf{R}}')' = \begin{bmatrix} R_{11}-\bar{R}_1 & R_{12}-\bar{R}_1 & \cdots & R_{1T}-\bar{R}_1 \\ R_{21}-\bar{R}_2 & R_{22}-\bar{R}_2 & \cdots & R_{2T}-\bar{R}_2 \\ \vdots & \vdots & \ddots & \vdots \\ R_{N1}-\bar{R}_N & R_{N2}-\bar{R}_N & \cdots & R_{NT}-\bar{R}_N \end{bmatrix} \tag{26}$$

It is clear that:

$$\begin{aligned}(\mathbf{R} - \mathbf{1}_T \bar{\boldsymbol{R}}^\top)^\top (\mathbf{R} - \mathbf{1}_T \bar{\boldsymbol{R}}^\top) &= \begin{bmatrix} R_{11}-\bar{R}_1 & R_{12}-\bar{R}_1 & \cdots & R_{1T}-\bar{R}_1 \\ R_{21}-\bar{R}_2 & R_{22}-\bar{R}_2 & \cdots & R_{2T}-\bar{R}_2 \\ \vdots & \vdots & \ddots & \vdots \\ R_{N1}-\bar{R}_N & R_{N2}-\bar{R}_N & \cdots & R_{NT}-\bar{R}_N \end{bmatrix} \begin{bmatrix} R_{11}-\bar{R}_1 & R_{21}-\bar{R}_2 & \cdots & R_{N1}-\bar{R}_N \\ R_{12}-\bar{R}_1 & R_{22}-\bar{R}_2 & \cdots & R_{N2}-\bar{R}_N \\ \vdots & \vdots & \ddots & \vdots \\ R_{1T}-\bar{R}_1 & R_{2T}-\bar{R}_2 & \cdots & R_{NT}-\bar{R}_N \end{bmatrix} \\ &= \begin{bmatrix} \sum_{t=1}^T (R_{1t}-\bar{R}_1)(R_{1t}-\bar{R}_1) & \sum_{t=1}^T (R_{1t}-\bar{R}_1)(R_{2t}-\bar{R}_2) & \cdots & \sum_{t=1}^T (R_{1t}-\bar{R}_1)(R_{Nt}-\bar{R}_N) \\ \sum_{t=1}^T (R_{2t}-\bar{R}_2)(R_{1t}-\bar{R}_1) & \sum_{t=1}^T (R_{2t}-\bar{R}_2)(R_{2t}-\bar{R}_2) & \cdots & \sum_{t=1}^T (R_{2t}-\bar{R}_2)(R_{Nt}-\bar{R}_N) \\ \vdots & \vdots & \ddots & \vdots \\ \sum_{t=1}^T (R_{Nt}-\bar{R}_N)(R_{1t}-\bar{R}_1) & \sum_{t=1}^T (R_{Nt}-\bar{R}_N)(R_{2t}-\bar{R}_2) & \cdots & \sum_{t=1}^T (R_{Nt}-\bar{R}_N)(R_{Nt}-\bar{R}_N) \end{bmatrix}. \end{aligned} \tag{27}$$

Then, it immediately follows that:

$$\mathbf{S} = \frac{1}{T}(\mathbf{R} - \mathbf{1}_T \bar{\boldsymbol{R}}^\top)^\top (\mathbf{R} - \mathbf{1}_T \bar{\boldsymbol{R}}^\top), \tag{28}$$

or

$$\mathbf{S} = \frac{1}{T}\tilde{\mathbf{R}}^\top \tilde{\mathbf{R}}, \tag{29}$$

where $\tilde{\mathbf{R}} = \mathbf{R} - \mathbf{1}_T \bar{\boldsymbol{R}}^\top$ is the **centered** (or **demeaned**) return matrix. This is the second expression of $\mathbf{S}$, which is in matrix form.

4.3 Third Expression

We next derive the last expression from the 2nd expression. Let's work with $(\mathbf{R} - \mathbf{1}_T\bar{\boldsymbol{R}}^\top)^\top(\mathbf{R} - \mathbf{1}_T\bar{\boldsymbol{R}}^\top)$. We can expand it:

$$\begin{aligned}(\mathbf{R} - \mathbf{1}_T\bar{\boldsymbol{R}}^\top)^\top(\mathbf{R} - \mathbf{1}_T\bar{\boldsymbol{R}}^\top) &= (\mathbf{R}^\top - (\mathbf{1}_T\bar{\boldsymbol{R}}^\top)^\top)(\mathbf{R} - \mathbf{1}_T\bar{\boldsymbol{R}}^\top) \\ &= \mathbf{R}^\top\mathbf{R} - \mathbf{R}^\top(\mathbf{1}_T\bar{\boldsymbol{R}}^\top) - (\mathbf{1}_T\bar{\boldsymbol{R}}^\top)^\top\mathbf{R} + (\mathbf{1}_T\bar{\boldsymbol{R}}^\top)^\top(\mathbf{1}_T\bar{\boldsymbol{R}}^\top).\end{aligned} \tag{30}$$

Let's simplify each term:

- $\mathbf{R}^\top\mathbf{R}$ remains as it is.
- $\mathbf{R}^\top(\mathbf{1}_T\bar{\boldsymbol{R}}^\top)$:

$$\mathbf{R}^\top(\mathbf{1}_T\bar{\boldsymbol{R}}^\top) = \mathbf{R}^\top\mathbf{1}_T\bar{\boldsymbol{R}}^\top. \tag{31}$$

  When we multiply $\mathbf{R}^\top$ (which is $N \times T$) by $\mathbf{1}_T$, we get a $N \times 1$ vector where each element is the sum of each column of $\mathbf{R}$. Therefore:

$$\mathbf{R}^\top\mathbf{1}_T = \begin{bmatrix} \sum_{t=1}^T R_{1t} \\ \sum_{t=1}^T R_{2t} \\ \vdots \\ \sum_{t=1}^T R_{Nt} \end{bmatrix}. \tag{32}$$

  This is a vector of the sum of the returns over time for each asset, which is $T\bar{\boldsymbol{R}}$. So:

$$\mathbf{R}^\top(\mathbf{1}_T\bar{\boldsymbol{R}}^\top) = (T\bar{\boldsymbol{R}})\bar{\boldsymbol{R}}^\top = T\bar{\boldsymbol{R}}\bar{\boldsymbol{R}}^\top. \tag{33}$$

- $(\mathbf{1}_T\bar{\boldsymbol{R}}^\top)^\top\mathbf{R}$: This term is similar to the previous one. Since $(\mathbf{1}_T\bar{\boldsymbol{R}}^\top)^\top = \bar{\boldsymbol{R}}(\mathbf{1}_T^\top)$, we get:

$$(\mathbf{1}_T\bar{\boldsymbol{R}}^\top)^\top\mathbf{R} = (\bar{\boldsymbol{R}}\mathbf{1}_T^\top)\mathbf{R} = \bar{\boldsymbol{R}}(\mathbf{1}_T^\top\mathbf{R}) = T\bar{\boldsymbol{R}}\bar{\boldsymbol{R}}^\top. \tag{34}$$

- $(\mathbf{1}_T\bar{\boldsymbol{R}}^\top)^\top(\mathbf{1}_T\bar{\boldsymbol{R}}^\top)$: Here, $(\mathbf{1}_T\bar{\boldsymbol{R}}^\top)^\top(\mathbf{1}_T\bar{\boldsymbol{R}}^\top)$ becomes:

$$(\bar{\boldsymbol{R}}\mathbf{1}_T^\top)(\mathbf{1}_T\bar{\boldsymbol{R}}^\top) = \bar{\boldsymbol{R}}(\mathbf{1}_T^\top\mathbf{1}_T)\bar{\boldsymbol{R}}^\top. \tag{35}$$

  Since $\mathbf{1}_T^\top\mathbf{1}_T$ is just the sum of ones, which is T, therefore:

$$(\bar{\boldsymbol{R}}\mathbf{1}_T^\top)(\mathbf{1}_T\bar{\boldsymbol{R}}^\top) = T\bar{\boldsymbol{R}}\bar{\boldsymbol{R}}^\top. \tag{36}$$

Combining all simplified terms, we have:

$$\begin{aligned}(\mathbf{R} - \mathbf{1}_T\bar{\boldsymbol{R}}^\top)^\top(\mathbf{R} - \mathbf{1}_T\bar{\boldsymbol{R}}^\top) &= \mathbf{R}^\top\mathbf{R} - T\bar{\boldsymbol{R}}\bar{\boldsymbol{R}}^\top - T\bar{\boldsymbol{R}}\bar{\boldsymbol{R}}^\top + T\bar{\boldsymbol{R}}\bar{\boldsymbol{R}}^\top \\ &= \mathbf{R}^\top\mathbf{R} - T\bar{\boldsymbol{R}}\bar{\boldsymbol{R}}^\top\end{aligned} \tag{37}$$

Dividing by T to obtain the 3rd expression of the sample covariance matrix:

$$\begin{aligned}\mathbf{S} &= \frac{1}{T}(\mathbf{R}-\mathbf{1}_T\bar{\boldsymbol{R}}^\top)^\top(\mathbf{R}-\mathbf{1}_T\bar{\boldsymbol{R}}^\top)\\ &= \frac{1}{T}\left(\mathbf{R}^\top\mathbf{R}-T\bar{\boldsymbol{R}}\bar{\boldsymbol{R}}^\top\right)\\ &= \frac{1}{T}\mathbf{R}^\top\mathbf{R}-\bar{\boldsymbol{R}}\bar{\boldsymbol{R}}^\top.\end{aligned} \tag{38}$$

As you can probably tell, this expression is pretty much the same as the second expression. They are both in matrix form, with the minor difference that the 2nd expression uses demeaned data, while the 3rd expression uses original data and the mean return vector. However, as we will see shortly, the 3rd expression is helpful for understanding recent advancement of PCA based latent factor models.

5 $(1/T)\tilde{\mathbf{R}}^\top\tilde{\mathbf{R}}$ and $(1/N)\tilde{\mathbf{R}}\tilde{\mathbf{R}}^\top$

Often times, instead of performing eigen-decomposition to the covariance matrix $(1/T)\tilde{\mathbf{R}}^\top\tilde{\mathbf{R}}$, we can also perform eigen-decomposition to the covariance matrix $(1/N)\tilde{\mathbf{R}}\tilde{\mathbf{R}}^\top$. In this section, we demonstrate how they are related.

Given a centered $T\times N$ data matrix $\tilde{\mathbf{R}}$, the SVD of $\tilde{\mathbf{R}}$ can be written as:

$$\tilde{\mathbf{R}}=\mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\top, \tag{39}$$

where $\mathbf{U}$ is a $T\times T$ orthogonal matrix (the left singular vectors of $\tilde{\mathbf{R}}$), $\mathbf{V}$ is a $N\times N$ orthogonal matrix (the right singular vectors of $\tilde{\mathbf{R}}$), and $\boldsymbol{\Sigma}$ is a $T\times N$ diagonal matrix with non-negative real numbers on the diagonal, denoted as σ_i, being the singular values of $\tilde{\mathbf{R}}$.

5.1 Connections between Eigenvalues

When we form the covariance matrices $\frac{1}{T}\tilde{\mathbf{R}}^\top\tilde{\mathbf{R}}$ and $\frac{1}{N}\tilde{\mathbf{R}}\tilde{\mathbf{R}}^\top$, the eigenvalues of these matrices are related to the singular values of $\tilde{\mathbf{R}}$.

1. Eigenvalues of $\frac{1}{T}\tilde{\mathbf{R}}^\top\tilde{\mathbf{R}}$:

$$\frac{1}{T}\tilde{\mathbf{R}}^\top\tilde{\mathbf{R}}=\frac{1}{T}\mathbf{V}\boldsymbol{\Sigma}^\top\boldsymbol{\Sigma}\mathbf{V}^\top=\mathbf{V}\left(\frac{\boldsymbol{\Sigma}^\top\boldsymbol{\Sigma}}{T}\right)\mathbf{V}^\top \tag{40}$$

   The eigenvalues of $\frac{1}{T}\tilde{\mathbf{R}}^\top\tilde{\mathbf{R}}$ are $\lambda_i=\frac{\sigma_i^2}{T}$.

2. Eigenvalues of $\frac{1}{N}\tilde{\mathbf{R}}\tilde{\mathbf{R}}^\top$:

$$\frac{1}{N}\tilde{\mathbf{R}}\tilde{\mathbf{R}}^\top=\frac{1}{N}\mathbf{U}\boldsymbol{\Sigma}\boldsymbol{\Sigma}^\top\mathbf{U}^\top=\mathbf{U}\left(\frac{\boldsymbol{\Sigma}\boldsymbol{\Sigma}^\top}{N}\right)\mathbf{U}^\top \tag{41}$$

The eigenvalues of $\frac{1}{N}\tilde{\mathbf{R}}\tilde{\mathbf{R}}^\top$ are $\mu_i = \frac{\sigma_i^2}{N}$.

From above, we see that the singular values σ_i are the non-negative square roots of the eigenvalues of $\frac{1}{T}\tilde{\mathbf{R}}^\top\tilde{\mathbf{R}}$ and $\frac{1}{N}\tilde{\mathbf{R}}\tilde{\mathbf{R}}^\top$:

$$\sigma_i = \sqrt{\lambda_i T} = \sqrt{\mu_i N}. \tag{42}$$

5.2 Connections between Eigenvectors

From $\frac{1}{T}\tilde{\mathbf{R}}^\top\tilde{\mathbf{R}} = \mathbf{V}\left(\frac{\mathbf{\Sigma}^\top\mathbf{\Sigma}}{T}\right)\mathbf{V}^\top$, it is easy to see that the columns of $\mathbf{V}$, $\boldsymbol{v}_i$, are the eigenvectors of $\frac{1}{T}\tilde{\mathbf{R}}^\top\tilde{\mathbf{R}}$. Similarly, from $\tilde{\mathbf{R}}\tilde{\mathbf{R}}^\top = \mathbf{U}\left(\frac{\mathbf{\Sigma}\mathbf{\Sigma}^\top}{N}\right)\mathbf{U}^\top$, it is easy to see that the columns of $\mathbf{U}$, $\boldsymbol{u}_i$, are the eigenvectors of $\frac{1}{N}\tilde{\mathbf{R}}\tilde{\mathbf{R}}^\top$.

Since $\tilde{\mathbf{R}}$ can be written as $\tilde{\mathbf{R}} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$, it implies that $\tilde{\mathbf{R}}\boldsymbol{v}_i = \sigma_i\boldsymbol{u}_i$ for each i. Then, by multiplying both sides of this equation by $\frac{1}{\sigma_i}$, we derive:

$$\boldsymbol{u}_i = \frac{1}{\sigma_i}\tilde{\mathbf{R}}\boldsymbol{v}_i. \tag{43}$$

This relationship between the eigenvectors, along with the connection between the eigenvalues, are crucial in demonstrating the equivalence of PCA performed on $\frac{1}{T}\tilde{\mathbf{R}}^\top\tilde{\mathbf{R}}$ and $\frac{1}{N}\tilde{\mathbf{R}}\tilde{\mathbf{R}}^\top$. Once you perform PCA on one of them, you will naturally derive the PCA for the other one.

For example, suppose we run PCA to $\frac{1}{N}\tilde{\mathbf{R}}\tilde{\mathbf{R}}^\top$ and derive its eigenvectors $\boldsymbol{u}_i$ and eigenvalues μ_i. Then, the eigenvectors and eigenvalues of $\frac{1}{T}\tilde{\mathbf{R}}^\top\tilde{\mathbf{R}}$ are:

- Eigenvalues: $\lambda_i = \dfrac{N}{T}\mu_i$,
- Eigenvectors: $\boldsymbol{v}_i = \dfrac{1}{\sigma_i}\tilde{\mathbf{R}}^\top\boldsymbol{u}_i$. (To see this, start with $\tilde{\mathbf{R}}\boldsymbol{v}_i = \sigma_i\boldsymbol{u}_i$, then $\tilde{\mathbf{R}}^\top\tilde{\mathbf{R}}\boldsymbol{v}_i = \sigma_i\tilde{\mathbf{R}}^\top\boldsymbol{u}_i$. Note that $\tilde{\mathbf{R}}^\top\tilde{\mathbf{R}} = \mathbf{V}\mathbf{\Sigma}^\top\mathbf{\Sigma}\mathbf{V}^\top$, and $\mathbf{V}\mathbf{\Sigma}^\top\mathbf{\Sigma}\mathbf{V}^\top\boldsymbol{v}_i = \sigma_i^2\boldsymbol{v}_i$. Hence, $\sigma_i^2\boldsymbol{v}_i = \sigma_i\tilde{\mathbf{R}}^\top\boldsymbol{u}_i$ or $\boldsymbol{v}_i = \frac{1}{\sigma_i}\tilde{\mathbf{R}}^\top\boldsymbol{u}_i$.)

5.3 Why Bother?

In empirical asset pricing, we often encounter situations where the number of assets N far exceeds the number of time periods T. For instance, in a cross-sectional analysis involving a large universe of stocks, N could be in the thousands, while T might only be in the hundreds or fewer. In such cases, performing PCA using the covariance matrix $\frac{1}{T}\tilde{\mathbf{R}}^\top\tilde{\mathbf{R}}$ becomes not only computationally intensive but also problematic in terms of matrix rank.

When $N > T$, the covariance matrix $\frac{1}{T}\tilde{\mathbf{R}}^\top\tilde{\mathbf{R}}$ is rank-deficient, meaning its rank cannot exceed T. This occurs because the data matrix $\tilde{\mathbf{R}}$, which is $T \times N$, has only T independent observations. As a result, the covariance matrix will have at most T non-zero eigenvalues, and the rest will be zero or close to zero. This rank deficiency implies that many principal components will correspond to zero variance, rendering them meaningless.

Therefore, although it's technically possible to compute PCA using the $N \times N$ covariance matrix, it's neither efficient nor useful in this context. Instead, a more practical approach is to work with the smaller $T \times T$ covariance matrix $\frac{1}{N}\tilde{\mathbf{R}}\tilde{\mathbf{R}}^\top$. This method leverages the properties of SVD to extract the same principal components but with significantly reduced computational burden. By focusing on the $T \times T$ matrix, we avoid the rank-deficiency issue and maintain the accuracy of PCA while improving efficiency.

6 Latest Development

6.1 Recap

As before, let $T \times N$ matrix $\mathbf{R}$ denote the (excess) returns (T periods and N assets), $T \times K$ (where $K \leq N$) matrix $\mathbf{F}$ denote the factor returns, and $N \times K$ matrix $\mathbf{\Lambda}$ denote the factor loadings. Let $\tilde{\mathbf{R}}$ and $\tilde{\mathbf{F}}$ represent the **demeaned** versions of $\mathbf{R}$ and $\mathbf{F}$, respectively.

From Section 3, we have learned that running PCA to the covariance matrix $\mathbf{S} = \frac{1}{T}\tilde{\mathbf{R}}^\top\tilde{\mathbf{R}} = \frac{1}{T}\mathbf{R}^\top\mathbf{R} - \bar{\boldsymbol{R}}\bar{\boldsymbol{R}}^\top$ is equivalent to solving the following least squares optimization problem (we omit the subscript K in $\mathbf{\Lambda}$ and $\mathbf{F}$ to simpify the notation):

$$\min_{\mathbf{\Lambda},\tilde{\mathbf{F}}} \frac{1}{NT}\sum_{n=1}^{N}\sum_{t=1}^{T}\left(\tilde{R}_{tn} - \tilde{\boldsymbol{F}}_t^\top \mathbf{\Lambda}_n\right)^2 \quad \text{with } \tilde{\mathbf{F}} = \tilde{\mathbf{R}}\mathbf{\Lambda}(\mathbf{\Lambda}^\top\mathbf{\Lambda})^{-1}, \tag{44}$$

where $\tilde{\boldsymbol{F}}_t^\top$ is the tth row of $\tilde{\mathbf{F}}$ (and hence $\tilde{\boldsymbol{F}}_t^\top$ is a $1 \times K$ row vector) and $\mathbf{\Lambda}_n^\top$ is the nth row of $\mathbf{\Lambda}$ (and hence $\mathbf{\Lambda}_n^\top$ is a $1 \times K$ row vector). This can be written in matrix form as:

$$\min_{\mathbf{\Lambda},\tilde{\mathbf{F}}} \frac{1}{NT}\|\tilde{\mathbf{R}} - \tilde{\mathbf{F}}\mathbf{\Lambda}^\top\|_F^2, \tag{45}$$

With the restriction $\tilde{\mathbf{F}} = \tilde{\mathbf{R}}\mathbf{\Lambda}(\mathbf{\Lambda}^\top\mathbf{\Lambda})^{-1}$, we can write $\tilde{\mathbf{R}} - \tilde{\mathbf{F}}\mathbf{\Lambda}^\top = \tilde{\mathbf{R}} - \tilde{\mathbf{R}}\mathbf{\Lambda}(\mathbf{\Lambda}^\top\mathbf{\Lambda})^{-1}\mathbf{\Lambda}^\top$, and simplify this expression by using the complement projection matrix $\mathbf{M} = \mathbf{I}_N - \mathbf{\Lambda}(\mathbf{\Lambda}^\top\mathbf{\Lambda})^{-1}\mathbf{\Lambda}^\top$:

$$\min_{\mathbf{\Lambda},\tilde{\mathbf{F}}} \frac{1}{NT}\|\tilde{\mathbf{R}}\mathbf{M}\|_F^2. \tag{46}$$

We can proceed by using trace operation (tr()) since $\|\mathbf{A}\|_F^2 = \text{tr}(\mathbf{A}^\top\mathbf{A}) = \text{tr}(\mathbf{A}\mathbf{A}^\top)$. Therefore, we can derive:

$$\frac{1}{NT}\|\tilde{\mathbf{R}}\mathbf{M}\|_F^2 = \frac{1}{NT}\text{tr}\left[(\tilde{\mathbf{R}}\mathbf{M})^\top(\tilde{\mathbf{R}}\mathbf{M})\right] = \frac{1}{NT}\text{tr}\left[\mathbf{M}\tilde{\mathbf{R}}^\top\tilde{\mathbf{R}}\mathbf{M}\right]. \tag{47}$$

This equation can be understood from right to left: minimizing $\frac{1}{NT}\text{tr}\left[\mathbf{M}\tilde{\mathbf{R}}^\top\tilde{\mathbf{R}}\mathbf{M}\right]$ is equivalent to minimizing $\frac{1}{NT}\|\tilde{\mathbf{R}}\mathbf{M}\|_F^2$. And $\frac{1}{NT}\|\tilde{\mathbf{R}}\mathbf{M}\|_F^2$ is merely the matrix form of the original objective function, where solving the original objective function is equivalent to performing PCA on the sample covariance matrix $\mathbf{S} = \frac{1}{T}\tilde{\mathbf{R}}^\top\tilde{\mathbf{R}} = \frac{1}{T}\mathbf{R}^\top\mathbf{R} - \bar{\boldsymbol{R}}\bar{\boldsymbol{R}}^\top$. At this point, note that the middle matrix $\tilde{\mathbf{R}}^\top\tilde{\mathbf{R}}$ in $\frac{1}{NT}\text{tr}\left[\mathbf{M}\tilde{\mathbf{R}}^\top\tilde{\mathbf{R}}\mathbf{M}\right]$ is nothing but a scaled version of $\mathbf{S}$. Therefore, the above equivalence

tells us that we can arbitrarily adjust the objective function of the optimization problem, and once it is transformed into the form of minimizing $\frac{1}{NT}\text{tr}\left[\mathbf{MAM}\right]$ through the trace operation, solving this adjusted optimization problem is equivalent to performing PCA on (a scaled version of) $\mathbf{A}$ (e.g. $\frac{1}{T}\mathbf{A}$). This greatly facilitates the process of solving the adjusted optimization problem. This equivalence-based solution method has also been adopted by the latest PCA methods proposed in recent years (Lettau and Pelger 2020, Bryzgalova et al. 2023).

Next, we first illustrate this with the Risk Premium PCA (RP-PCA) proposed by Lettau and Pelger (2020).

6.2 Risk Premium PCA

Now, instead of the optimization problem that corresponds to the traditional PCA, consider the following new optimization problem that minimizes a weighted sum of both time-series variation of returns and the cross-sectional pricing errors:

$$\min_{\mathbf{\Lambda},\mathbf{F}} \frac{1}{NT}\sum_{n=1}^{N}\sum_{t=1}^{T}\left(R_{tn}-\boldsymbol{F}_t^\top\mathbf{\Lambda}_n\right)^2+\frac{\gamma}{N}\sum_{n=1}^{N}\left(\bar{R}_n-\mathbf{\Lambda}_n^\top\bar{\boldsymbol{F}}\right)^2 \quad \text{with } \mathbf{F}=\mathbf{R}\mathbf{\Lambda}(\mathbf{\Lambda}^\top\mathbf{\Lambda})^{-1}, \tag{48}$$

where $\boldsymbol{F}_t^\top$ is the tth row of $\mathbf{F}$, $\mathbf{\Lambda}_n^\top$ is the nth row of $\mathbf{\Lambda}$, $\bar{R}_n$ is the time-series average return of asset n, $\bar{\boldsymbol{F}}$ represents the time-series average return of all K factors (and hence it is a $K\times 1$ vector), and finally γ is an coefficient that determines the relative strengths of the two components. (I strongly encourage you to check the dimensions of the vectors and matrices yourself to ensure you understand them!!!)

In this new objective function, we consider not only the time-series variation of returns (1st term), but also the cross-sectional pricing errors (2nd term). Now, let's explore further to see the beautiful idea behind solving this optimization problem. To start with, we rewrite the objective function in matrix form (check this by yourself!!!):

$$\frac{1}{NT}\|\mathbf{R}-\mathbf{F}\mathbf{\Lambda}^\top\|_F^2+\frac{\gamma}{N}\|\bar{\boldsymbol{R}}-\mathbf{\Lambda}\bar{\boldsymbol{F}}\|_F^2. \tag{49}$$

Let's look at the first term. Again, with the restriction $\mathbf{F}=\mathbf{R}\mathbf{\Lambda}(\mathbf{\Lambda}^\top\mathbf{\Lambda})^{-1}$, we can write $\mathbf{R}-\mathbf{F}\mathbf{\Lambda}^\top=\mathbf{R}-\mathbf{R}\mathbf{\Lambda}(\mathbf{\Lambda}^\top\mathbf{\Lambda})^{-1}\mathbf{\Lambda}^\top$. Again, we simplify this expression by using $\mathbf{M}=\mathbf{I}_N-\mathbf{\Lambda}(\mathbf{\Lambda}^\top\mathbf{\Lambda})^{-1}\mathbf{\Lambda}^\top$, and therefore $\mathbf{R}-\mathbf{F}\mathbf{\Lambda}^\top=\mathbf{R}\mathbf{M}$.

For the second term, first, note the following holds true:

$$\begin{aligned}\bar{\boldsymbol{R}}&=\frac{1}{T}\mathbf{R}^\top\mathbf{1}_T,\\ \bar{\boldsymbol{F}}&=\frac{1}{T}\mathbf{F}^\top\mathbf{1}_T,\end{aligned} \tag{50}$$

where $\mathbf{1}_T$ is a $T \times 1$ vector of ones. Therefore, $\bar{\boldsymbol{R}} - \boldsymbol{\Lambda}\bar{\boldsymbol{F}}$ can be written as:

$$\begin{aligned} & \frac{1}{T}\mathbf{R}^\top \mathbf{1}_T - \boldsymbol{\Lambda}\frac{1}{T}\mathbf{F}^\top \mathbf{1}_T \\ = & \frac{1}{T}\left(\mathbf{R}^\top - \boldsymbol{\Lambda}\mathbf{F}^\top\right)\mathbf{1}_T \\ = & \frac{1}{T}\left(\mathbf{R} - \mathbf{F}\boldsymbol{\Lambda}^\top\right)^\top \mathbf{1}_T \\ = & \frac{1}{T}\left(\mathbf{RM}\right)^\top \mathbf{1}_T \\ = & \frac{1}{T}\mathbf{M}\mathbf{R}^\top \mathbf{1}_T. \end{aligned} \tag{51}$$

The last equality above uses the fact that $\mathbf{M}$ is symmetric. Now, the objective function becomes:

$$\frac{1}{NT}\|\mathbf{RM}\|_F^2 + \frac{\gamma}{N}\|\frac{1}{T}\mathbf{M}\mathbf{R}^\top \mathbf{1}_T\|_F^2. \tag{52}$$

As before, we proceed by using trace operation:

$$\begin{aligned} & \frac{1}{NT}\|\mathbf{RM}\|_F^2 + \frac{\gamma}{N}\|\frac{1}{T}\mathbf{M}\mathbf{R}^\top \mathbf{1}_T\|_F^2 \\ = & \frac{1}{NT}\text{tr}\left[(\mathbf{RM})^\top(\mathbf{RM})\right] + \frac{\gamma}{N}\text{tr}\left[\left(\frac{1}{T}\mathbf{M}\mathbf{R}^\top \mathbf{1}_T\right)\left(\frac{1}{T}\mathbf{M}\mathbf{R}^\top \mathbf{1}_T\right)^\top\right] \\ = & \frac{1}{NT}\text{tr}\left[(\mathbf{RM})^\top(\mathbf{RM})\right] + \frac{\gamma}{N}\text{tr}\left[\left(\mathbf{M}\bar{\boldsymbol{R}}\right)\left(\mathbf{M}\bar{\boldsymbol{R}}\right)^\top\right] \\ = & \frac{1}{NT}\text{tr}\left[\mathbf{M}\mathbf{R}^\top\mathbf{R}\mathbf{M}\right] + \frac{\gamma}{N}\text{tr}\left[\mathbf{M}\bar{\boldsymbol{R}}\bar{\boldsymbol{R}}^\top\mathbf{M}\right] \\ = & \text{tr}\left[\frac{1}{NT}\mathbf{M}\mathbf{R}^\top\mathbf{R}\mathbf{M} + \frac{\gamma}{N}\mathbf{M}\bar{\boldsymbol{R}}\bar{\boldsymbol{R}}^\top\mathbf{M}\right] \\ = & \text{tr}\left[\frac{1}{NT}\left(\mathbf{M}\mathbf{R}^\top\mathbf{R}\mathbf{M} + \gamma T\mathbf{M}\bar{\boldsymbol{R}}\bar{\boldsymbol{R}}^\top\mathbf{M}\right)\right] \\ = & \frac{1}{NT}\text{tr}\left[\mathbf{M}\left(\mathbf{R}^\top\mathbf{R} + \gamma T\bar{\boldsymbol{R}}\bar{\boldsymbol{R}}^\top\right)\mathbf{M}\right]. \end{aligned} \tag{53}$$

Compare this new objective function in terms of trace with the one corresponding to the traditional PCA, it is immediately clear that the middle matrix is $\mathbf{R}^\top\mathbf{R} + \gamma T\bar{\boldsymbol{R}}\bar{\boldsymbol{R}}^\top$, and that solving the new optimization problem is equivalent to conducting eigen-decomposition to the following matrix:

$$\boldsymbol{\Sigma}_{\text{RP}} = \frac{1}{T}\left[\mathbf{R}^\top\mathbf{R} + \gamma T\bar{\boldsymbol{R}}\bar{\boldsymbol{R}}^\top\right] = \frac{1}{T}\mathbf{R}^\top\mathbf{R} + \gamma\bar{\boldsymbol{R}}\bar{\boldsymbol{R}}^\top. \tag{54}$$

This is called the **Risk Premium PCA** (RP-PCA) since the objective function considers both time-series and cross-sectional information of asset returns. It is easy to see that, when $\gamma = -1$, the $\boldsymbol{\Sigma}_{\text{RP}}$ reduces to the sample covariance matrix, and RP-PCA coincides with traditional PCA.

6.3 PCA with Economic Targets

Let's move one step further (and this is a big step) and look at a more general approach than RP-PCA. Bryzgalova et al. (2023) proposed a general PCA framework that allows users to incorporate cross-sectional, time-series, or both information into the objective function. (BTW, both Markus

Pelger and Svetlana Bryzgalova are on my list of favorite Finance professors.) I will talk about how to integrate cross-sectional information into traditional PCA in detail and leave the time-series part to you.

Extensive research in empirical asset pricing has demonstrated that portfolios constructed by sorting based on firm characteristics exhibit monotonic returns (e.g., Book-to-Market ratio, size, profitability, etc.). Suppose we consider L such portfolios and let the $N \times L$ matrix $\mathbf{W}^{XS}$ represent the weights of stocks within these portfolios. Then, the returns of these portfolios can be expressed as $\mathbf{R}^{XS} = \mathbf{R}\mathbf{W}^{XS}$, where $\mathbf{R}^{XS}$ is a $T \times L$ matrix.

Each of the L portfolios is merely a linear combination of the N stocks. Therefore, their factor exposures are also a weighted average of the individual stock factor exposures. In our estimated model, the $N \times K$ matrix $\mathbf{\Lambda}$ represents the exposures of individual stocks to K latent factors. Consequently, it is evident that the factor exposures of these additional L portfolios to the K latent factors are given by:

$$\underbrace{\mathbf{\Lambda}^{XS}}_{L\times K} = \underbrace{(\mathbf{W}^{XS})^\top}_{L\times N}\underbrace{(\mathbf{\Lambda})}_{N\times K}. \tag{55}$$

We want the factor exposures of these portfolios to the latent factors, $\mathbf{\Lambda}^{XS}$, and their returns, $\mathbf{R}^{XS}$, to exhibit high correlation. This correlation will guide the estimation process of the latent factors, deviating from the traditional PCA approach by incorporating the cross-sectional information contained in these L portfolios (reflected in $\mathbf{W}^{XS}$). To this end, consider the following least squares problem:

$$\min_{\mathbf{\Lambda},\mathbf{F}} \frac{1}{NT}\sum_{n=1}^{N}\sum_{t=1}^{T}\left(R_{tn} - \mathbf{F}_t^\top \mathbf{\Lambda}_n\right)^2 + \frac{\gamma}{LT}\sum_{l=1}^{L}\sum_{t=1}^{T}\left(R_{tl}^{XS} - \mathbf{F}_t^\top(\mathbf{\Lambda}_l^{XS})\right)^2, \tag{56}$$

where $(\mathbf{\Lambda}_l^{XS})^\top$ is the lth row of $\mathbf{\Lambda}^{XS}$, and γ is an coefficient that determines the relative strengths of the two components. In this objective function, the first term corresponds to the traditional PCA, while the second term represents the new cross-sectional information. In matrix form, the second term becomes:

$$\frac{\gamma}{LT}\sum_{l=1}^{L}\sum_{t=1}^{T}\left(R_{tl}^{XS} - \mathbf{F}_t^\top(\mathbf{\Lambda}_l^{XS})\right)^2 = \frac{\gamma}{LT}\|\mathbf{R}\mathbf{W}^{XS} - \mathbf{F}\mathbf{\Lambda}^\top\mathbf{W}^{XS}\|_F^2. \tag{57}$$

When introducing the additional L portfolios, we do not impose stringent constraints such as requiring them to be independent (although portfolios constructed based on different firm characteristics generally have low correlations). Therefore, these portfolios may contain some redundant information. To address this, we introduce the $N \times N$ cross-sectional projection matrix $\mathbf{P}$:

$$\mathbf{P} = \mathbf{W}^{XS}\left((\mathbf{W}^{XS})^\top(\mathbf{W}^{XS})\right)^{-1}(\mathbf{W}^{XS})^\top. \tag{58}$$

Using this projection matrix, we can transform the aforementioned second term to :

$$\frac{\gamma}{LT}\|\mathbf{R}\mathbf{P}-\mathbf{F}\mathbf{\Lambda}^\top\mathbf{P}\|_F^2. \tag{59}$$

The second term can be understood as projecting the returns of the original N stocks onto the column space of $\mathbf{W}^{XS}$ and minimizing the pricing error of these L portfolios after the projection. Therefore, instead of using $\mathbf{W}^{XS}$, we use the projection matrix $\mathbf{P}$.

Finally, the objective function in matrix form becomes:

$$\frac{1}{NT}\|\mathbf{R}-\mathbf{F}\mathbf{\Lambda}^\top\|_F^2+\frac{\gamma}{LT}\|\mathbf{R}\mathbf{P}-\mathbf{F}\mathbf{\Lambda}^\top\mathbf{P}\|_F^2. \tag{60}$$

By changing the coefficient γ, we can achieve a balance between the two toward optimal out-of-sample performance of the estimated latent factors. Next, as you might have guessed, to solve this least squares, I aim to derive from this matrix-form objective function an equivalent matrix for which we can perform PCA. To do so, I will introduce a more general and powerful approach (much better than the method in Section 6.2 that introduces $\mathbf{M}$).

I am indebt to Prof. WANG, Xi of Peking University (School of Economics) for his insightful discussion with me regarding this approach. Prof. Wang is a long-term collaborator of mine. Highly recommended! BTW, he has an asset pricing book forthcoming, which I shall recommend without any hesitation. Stay tuned.

Rewirte the objective function as:

$$\frac{1}{NT}\left\{\|\mathbf{R}-\mathbf{F}\mathbf{\Lambda}^\top\|_F^2+\frac{\gamma N}{L}\|\mathbf{R}\mathbf{P}-\mathbf{F}\mathbf{\Lambda}^\top\mathbf{P}\|_F^2\right\}. \tag{61}$$

Again, using trace operation, the above objective function is propotional to the following trace:

$$\begin{aligned}&\mathrm{tr}\left[(\mathbf{R}-\mathbf{F}\mathbf{\Lambda}^\top)^\top(\mathbf{R}-\mathbf{F}\mathbf{\Lambda}^\top)\right]+\mathrm{tr}\left[\frac{\gamma N}{L}\left[(\mathbf{R}-\mathbf{F}\mathbf{\Lambda}^\top)\mathbf{P}\right]^\top\left[(\mathbf{R}-\mathbf{F}\mathbf{\Lambda}^\top)\mathbf{P}\right]\right]\\ =\;&\mathrm{tr}\left[(\mathbf{R}^\top-\mathbf{\Lambda}\mathbf{F}^\top)(\mathbf{R}-\mathbf{F}\mathbf{\Lambda}^\top)\right]+\frac{\gamma N}{L}\mathrm{tr}\left[\mathbf{P}^\top(\mathbf{R}^\top-\mathbf{\Lambda}\mathbf{F}^\top)(\mathbf{R}-\mathbf{F}\mathbf{\Lambda}^\top)\mathbf{P}\right].\end{aligned} \tag{62}$$

Using the cyclic property (I will make use of this property many times!) of trace operation and the fact that $\mathbf{P}\mathbf{P}^\top=\mathbf{P}$ (since $\mathbf{P}$ is a projection matrix) to the second term:

$$\begin{aligned}&\mathrm{tr}\left[(\mathbf{R}^\top-\mathbf{\Lambda}\mathbf{F}^\top)(\mathbf{R}-\mathbf{F}\mathbf{\Lambda}^\top)\right]+\frac{\gamma N}{L}\mathrm{tr}\left[\mathbf{P}^\top(\mathbf{R}^\top-\mathbf{\Lambda}\mathbf{F}^\top)(\mathbf{R}-\mathbf{F}\mathbf{\Lambda}^\top)\mathbf{P}\right]\\ =\;&\mathrm{tr}\left[(\mathbf{R}^\top-\mathbf{\Lambda}\mathbf{F}^\top)(\mathbf{R}-\mathbf{F}\mathbf{\Lambda}^\top)\right]+\frac{\gamma N}{L}\mathrm{tr}\left[(\mathbf{R}^\top-\mathbf{\Lambda}\mathbf{F}^\top)(\mathbf{R}-\mathbf{F}\mathbf{\Lambda}^\top)\underbrace{\mathbf{P}\mathbf{P}^\top}_{\mathbf{P}}\right]\\ =\;&\mathrm{tr}\left[(\mathbf{R}^\top-\mathbf{\Lambda}\mathbf{F}^\top)(\mathbf{R}-\mathbf{F}\mathbf{\Lambda}^\top)\right]+\frac{\gamma N}{L}\mathrm{tr}\left[(\mathbf{R}^\top-\mathbf{\Lambda}\mathbf{F}^\top)(\mathbf{R}-\mathbf{F}\mathbf{\Lambda}^\top)\mathbf{P}\right]\\ =\;&\mathrm{tr}\left[(\mathbf{R}^\top-\mathbf{\Lambda}\mathbf{F}^\top)(\mathbf{R}-\mathbf{F}\mathbf{\Lambda}^\top)+\frac{\gamma N}{L}(\mathbf{R}^\top-\mathbf{\Lambda}\mathbf{F}^\top)(\mathbf{R}-\mathbf{F}\mathbf{\Lambda}^\top)\mathbf{P}\right]\\ =\;&\mathrm{tr}\left[(\mathbf{R}^\top-\mathbf{\Lambda}\mathbf{F}^\top)(\mathbf{R}-\mathbf{F}\mathbf{\Lambda}^\top)\left(\mathbf{I}_N+\frac{\gamma N}{L}\mathbf{P}\right)\right].\end{aligned} \tag{63}$$

Next, since $\mathbf{I}_N$ is identity matrix and $\mathbf{P}$ is a projection matrix, we write $(\mathbf{I}_N + \frac{\gamma N}{L}\mathbf{P})$ as $(\mathbf{I}_N + \frac{\gamma N}{L}\mathbf{P})^{0.5}(\mathbf{I}_N + \frac{\gamma N}{L}\mathbf{P})^{0.5}$, and $(\mathbf{I}_N + \frac{\gamma N}{L}\mathbf{P})^{0.5}$ has nice property (e.g., it is symmetric; check this by yourself!). Then, our objective function becomes:

$$\begin{aligned} & \text{tr}\left[(\mathbf{R}^\top - \mathbf{\Lambda}\mathbf{F}^\top)(\mathbf{R} - \mathbf{F}\mathbf{\Lambda}^\top)\left(\mathbf{I}_N + \frac{\gamma N}{L}\mathbf{P}\right)\right] \\ = & \text{tr}\left[(\mathbf{R}^\top - \mathbf{\Lambda}\mathbf{F}^\top)(\mathbf{R} - \mathbf{F}\mathbf{\Lambda}^\top)\left(\mathbf{I}_N + \frac{\gamma N}{L}\mathbf{P}\right)^{0.5}\left(\mathbf{I}_N + \frac{\gamma N}{L}\mathbf{P}\right)^{0.5}\right] \\ = & \text{tr}\left[\left(\mathbf{I}_N + \frac{\gamma N}{L}\mathbf{P}\right)^{0.5}(\mathbf{R}^\top - \mathbf{\Lambda}\mathbf{F}^\top)(\mathbf{R} - \mathbf{F}\mathbf{\Lambda}^\top)\left(\mathbf{I}_N + \frac{\gamma N}{L}\mathbf{P}\right)^{0.5}\right]. \end{aligned} \tag{64}$$

To proceed, let $\hat{\mathbf{I}} = \left(\mathbf{I}_N + \frac{\gamma N}{L}\mathbf{P}\right)^{0.5}$, then:

$$\begin{aligned} & \text{tr}\left[\hat{\mathbf{I}}(\mathbf{R}^\top - \mathbf{\Lambda}\mathbf{F}^\top)(\mathbf{R} - \mathbf{F}\mathbf{\Lambda}^\top)\hat{\mathbf{I}}\right] \\ = & \text{tr}\left[(\hat{\mathbf{I}}\mathbf{R}^\top - \hat{\mathbf{I}}\mathbf{\Lambda}\mathbf{F}^\top)(\mathbf{R}\hat{\mathbf{I}} - \mathbf{F}\mathbf{\Lambda}^\top\hat{\mathbf{I}})\right] \\ = & \text{tr}\left[\hat{\mathbf{I}}\mathbf{R}^\top\mathbf{R}\hat{\mathbf{I}} - \hat{\mathbf{I}}\mathbf{R}^\top\mathbf{F}\mathbf{\Lambda}^\top\hat{\mathbf{I}} - \hat{\mathbf{I}}\mathbf{\Lambda}\mathbf{F}^\top\mathbf{R}\hat{\mathbf{I}} + \hat{\mathbf{I}}\mathbf{\Lambda}\mathbf{F}^\top\mathbf{F}\mathbf{\Lambda}^\top\hat{\mathbf{I}}\right] \\ = & \text{tr}\left[\hat{\mathbf{I}}\mathbf{R}^\top\mathbf{R}\hat{\mathbf{I}}\right] - \text{tr}\left[\hat{\mathbf{I}}\mathbf{R}^\top\mathbf{F}\mathbf{\Lambda}^\top\hat{\mathbf{I}}\right] - \text{tr}\left[\hat{\mathbf{I}}\mathbf{\Lambda}\mathbf{F}^\top\mathbf{R}\hat{\mathbf{I}}\right] + \text{tr}\left[\hat{\mathbf{I}}\mathbf{\Lambda}\mathbf{F}^\top\mathbf{F}\mathbf{\Lambda}^\top\hat{\mathbf{I}}\right]. \end{aligned} \tag{65}$$

In this expression above, it is easy to show that the 2nd and 3rd terms are the same:

$$\text{tr}[(\hat{\mathbf{I}}\mathbf{R}^\top)(\mathbf{F}\mathbf{\Lambda}^\top\hat{\mathbf{I}})] = \text{tr}[(\mathbf{R}\hat{\mathbf{I}})^\top(\mathbf{F}\mathbf{\Lambda}^\top\hat{\mathbf{I}})] = \text{tr}[(\mathbf{F}\mathbf{\Lambda}^\top\hat{\mathbf{I}})^\top(\mathbf{R}\hat{\mathbf{I}})] = \text{tr}[(\hat{\mathbf{I}}\mathbf{\Lambda}\mathbf{F}^\top)(\mathbf{R}\hat{\mathbf{I}})], \tag{66}$$

where in the 2nd equality I use the fact that $\text{tr}(\mathbf{A}^\top\mathbf{B}) = \text{tr}(\mathbf{B}^\top\mathbf{A})$; besides, I have used the fact that $\hat{\mathbf{I}}$ is symmetric so that $\hat{\mathbf{I}} = \hat{\mathbf{I}}^\top$ to simplfy the notation. Then, we can proceed as follows:

$$\begin{aligned} & \text{tr}\left[\hat{\mathbf{I}}\mathbf{R}^\top\mathbf{R}\hat{\mathbf{I}}\right] - \text{tr}\left[\hat{\mathbf{I}}\mathbf{R}^\top\mathbf{F}\mathbf{\Lambda}^\top\hat{\mathbf{I}}\right] - \text{tr}\left[\hat{\mathbf{I}}\mathbf{\Lambda}\mathbf{F}^\top\mathbf{R}\hat{\mathbf{I}}\right] + \text{tr}\left[\hat{\mathbf{I}}\mathbf{\Lambda}\mathbf{F}^\top\mathbf{F}\mathbf{\Lambda}^\top\hat{\mathbf{I}}\right] \\ = & \text{tr}\left[\hat{\mathbf{I}}\mathbf{R}^\top\mathbf{R}\hat{\mathbf{I}}\right] - 2\text{tr}\left[\hat{\mathbf{I}}\mathbf{R}^\top\mathbf{F}\mathbf{\Lambda}^\top\hat{\mathbf{I}}\right] + \text{tr}\left[\hat{\mathbf{I}}\mathbf{\Lambda}\mathbf{F}^\top\mathbf{F}\mathbf{\Lambda}^\top\hat{\mathbf{I}}\right] \\ = & \text{tr}\left[\hat{\mathbf{I}}\mathbf{R}^\top\mathbf{R}\hat{\mathbf{I}} - 2\hat{\mathbf{I}}\mathbf{R}^\top\mathbf{F}\mathbf{\Lambda}^\top\hat{\mathbf{I}} + \hat{\mathbf{I}}\mathbf{\Lambda}\mathbf{F}^\top\mathbf{F}\mathbf{\Lambda}^\top\hat{\mathbf{I}}\right]. \end{aligned} \tag{67}$$

Define:

$$\begin{aligned} \underbrace{\hat{\mathbf{\Lambda}}}_{N\times K} &= \hat{\mathbf{I}}\mathbf{\Lambda}, \\ \hat{\mathbf{R}} &= \mathbf{R}\hat{\mathbf{I}}. \end{aligned} \tag{68}$$

Then:

$$\begin{aligned} & \text{tr}\left[\hat{\mathbf{I}}\mathbf{R}^\top\mathbf{R}\hat{\mathbf{I}} - 2\hat{\mathbf{I}}\mathbf{R}^\top\mathbf{F}\mathbf{\Lambda}^\top\hat{\mathbf{I}} + \hat{\mathbf{I}}\mathbf{\Lambda}\mathbf{F}^\top\mathbf{F}\mathbf{\Lambda}^\top\hat{\mathbf{I}}\right] \\ = & \text{tr}\left[\underbrace{\hat{\mathbf{R}}^\top}_{N\times T}\underbrace{\hat{\mathbf{R}}}_{T\times N} - 2\underbrace{\hat{\mathbf{R}}^\top}_{N\times T}\underbrace{\mathbf{F}}_{T\times K}\underbrace{\hat{\mathbf{\Lambda}}^\top}_{K\times N} + \underbrace{\hat{\mathbf{\Lambda}}}_{N\times K}\underbrace{\mathbf{F}^\top}_{K\times T}\underbrace{\mathbf{F}}_{T\times K}\underbrace{\hat{\mathbf{\Lambda}}^\top}_{K\times N}\right] \\ = & \text{tr}\left[\left(\hat{\mathbf{R}} - \mathbf{F}\hat{\mathbf{\Lambda}}^\top\right)^\top\left(\hat{\mathbf{R}} - \mathbf{F}\hat{\mathbf{\Lambda}}^\top\right)\right] \\ = & \|\hat{\mathbf{R}} - \mathbf{F}\hat{\mathbf{\Lambda}}^\top\|_F^2. \end{aligned} \tag{69}$$

So, after all these derivation, we find that the original objective function is equivalent to the

following:

$$\min \|\hat{\mathbf{R}} - \mathbf{F}\hat{\mathbf{\Lambda}}^\top\|_F^2. \tag{70}$$

To solve this least squares, we can apply eigen-decomposition to the following $T \times T$ matrix:

$$\hat{\mathbf{R}}\hat{\mathbf{R}}^\top = \mathbf{R}\hat{\mathbf{I}}\hat{\mathbf{I}}\mathbf{R}^\top = \mathbf{R}\left(\mathbf{I}_N + \frac{\gamma N}{L}\mathbf{P}\right)\mathbf{R}^\top. \tag{71}$$

We are done! Once we back out the first K eigenvectors from eigen-decomposition, they will serve as the factor returns of first K factors, and we can derive factor exposures as:

$$\hat{\mathbf{\Lambda}} = \mathbf{R}^\top\hat{\mathbf{F}}. \tag{72}$$

By exactly the same technique, you can derive the time-series part of Bryzgalova et al. (2023). I strongly encourage you to give it a try to master this elegant technique.

6.4 Instrumented PCA

6.4.1 Motivation and Model

The methods introduced above for estimating latent factors through PCA, whether traditional PCA, RP-PCA, or PCA with economic targets, are essentially static models. In these models, the estimated factor exposures $\mathbf{\Lambda}$ do not vary over time. However, it is evident that the various characteristics of companies change over time. For instance, a high-growth company may lose its growth potential in its later stages, or a small-cap company may develop into a large-cap company over time. This means that a company's stock exposure to given factors is likely to be time-varying rather than static. Nevertheless, the aforementioned PCA methods do not account for this. Another shortcoming of the above PCA methods (except for PCA with economic targets) is that they mainly rely on the information contained in returns and do not explicitly incorporate other information related to future stock returns, such as firm characteristics. The exception of PCA with economic targets is because it can introduce some firm characteristics through cross-sectional or time-series information.

Given the above discussion, a natural extension is to consider dynamic factor models (i.e., where individual stock factor exposures change over time) and incorporate information other than returns into the model. A representative work in this area is the Instrumented PCA (IPCA) model proposed by Kelly, Pruitt, and Su (2019). (Incidentally, Bryan Kelly is also on my list of favorite finance professors.)

For time t, consider the following latent factor model:

$$\underbrace{\boldsymbol{r}_{t+1}}_{N\times 1} = \underbrace{\mathbf{\Lambda}_t}_{N\times K}\underbrace{\boldsymbol{f}_{t+1}}_{K\times 1} + \boldsymbol{e}_{t+1}. \tag{73}$$

Kelly, Pruitt, and Su (2019) considered conditional factor loadings as functions of L observable firm

characteristics (known as *instruments*):

$$\underbrace{\mathbf{\Lambda}_t}_{N\times K} = \underbrace{\mathbf{Z}_t}_{N\times L}\underbrace{\mathbf{\Gamma}}_{L\times K}, \tag{74}$$

where $\mathbf{Z}_t$ stores the firm characteristics data of time t, and $\mathbf{\Gamma}$ maps $\mathbf{Z}_t$ to $\mathbf{\Lambda}_t$; $\mathbf{\Gamma}$ needs to be estimated. Combining them together gives:

$$\boldsymbol{r}_{t+1} = \mathbf{Z}_t\mathbf{\Gamma}\boldsymbol{f}_{t+1} + \boldsymbol{e}_{t+1}. \tag{75}$$

The advantage of this model is that $\mathbf{\Gamma}$ does not have a subscript t, implying that the model assumes it does not change over time. In other words, all variations in individual stock factor exposures are captured through the time-varying firm characteristics $\mathbf{Z}_t$. In this model, $\mathbf{\Gamma}$ is crucial as it allows the model to choose a reduced-rank combination of the L candidate characteristics that best describe the factor loading structure. Since $\mathbf{Z}_t$ is given, the goal of estimating this model is to find $\mathbf{\Gamma}$ and $\mathbf{F}$ that minimize the following error:

$$\min_{\mathbf{\Gamma},\mathbf{F}} \sum_{t=1}^{T-1} \left(\boldsymbol{r}_{t+1} - \mathbf{Z}_t\mathbf{\Gamma}\boldsymbol{f}_{t+1}\right)^\top \left(\boldsymbol{r}_{t+1} - \mathbf{Z}_t\mathbf{\Gamma}\boldsymbol{f}_{t+1}\right). \tag{76}$$

6.4.2 Estimation

The estimation is similar to traditional PCA, i.e., we can optimize the least squares to find the latent factors $\{\boldsymbol{f}_t\}$ and loadings $\mathbf{\Gamma}$. Since the model is constructed by adding instruments into a PCA framework, the authors term it the instrumented PCA or IPCA model. Specifically:

- given estimate of $\mathbf{\Gamma}$, we have $\boldsymbol{f}_{t+1} = \left(\mathbf{\Gamma}^\top\mathbf{Z}_t^\top\mathbf{Z}_t\mathbf{\Gamma}\right)^{-1}\mathbf{\Gamma}^\top\mathbf{Z}_t^\top\boldsymbol{r}_{t+1}$;
- given estimates of the series $\{\mathbf{f}_t\}, t = 1, \cdots, T-1$, we have

$$\text{vec}(\mathbf{\Gamma}^\top) = \left(\sum_{t=1}^{T-1}\mathbf{Z}_t^\top\mathbf{Z}_t \otimes \boldsymbol{f}_{t+1}\boldsymbol{f}_{t+1}^\top\right)^{-1}\left(\sum_{t=1}^{T-1}\left[\mathbf{Z}_t \otimes \boldsymbol{f}_{t+1}^\top\right]^\top\boldsymbol{r}_{t+1}\right). \tag{77}$$

  where $\otimes$ is Kronecker product.

How do we derive the expression for $\text{vec}(\mathbf{\Gamma}^\top)$? For a given t, we want to minimize

$$\begin{aligned} & \left(\boldsymbol{r}_{t+1} - \mathbf{Z}_t\mathbf{\Gamma}\boldsymbol{f}_{t+1}\right)^\top\left(\boldsymbol{r}_{t+1} - \mathbf{Z}_t\mathbf{\Gamma}\boldsymbol{f}_{t+1}\right) \\ = \; & \left(\boldsymbol{r}_{t+1}^\top - \boldsymbol{f}_{t+1}^\top\mathbf{\Gamma}^\top\mathbf{Z}_t^\top\right)\left(\boldsymbol{r}_{t+1} - \mathbf{Z}_t\mathbf{\Gamma}\mathbf{f}_{t+1}\right) \\ = \; & \boldsymbol{r}_{t+1}^\top\boldsymbol{r}_{t+1} - \boldsymbol{f}_{t+1}^\top\mathbf{\Gamma}^\top\mathbf{Z}_t^\top\boldsymbol{r}_{t+1} - \boldsymbol{r}_{t+1}^\top\mathbf{Z}_t\mathbf{\Gamma}\mathbf{f}_{t+1} + \boldsymbol{f}_{t+1}^\top\mathbf{\Gamma}^\top\mathbf{Z}_t^\top\mathbf{Z}_t\mathbf{\Gamma}\boldsymbol{f}_{t+1}. \end{aligned} \tag{78}$$

The first term above is irrelevant and using the trace operator, we have:

$$\begin{aligned} & -\text{tr}\left[\mathbf{f}_{t+1}^\top\mathbf{\Gamma}^\top\mathbf{Z}_t^\top\boldsymbol{r}_{t+1}\right] - \text{tr}\left[\boldsymbol{r}_{t+1}^\top\mathbf{Z}_t\mathbf{\Gamma}\boldsymbol{f}_{t+1}\right] + \text{tr}\left[\boldsymbol{f}_{t+1}^\top\mathbf{\Gamma}^\top\mathbf{Z}_t^\top\mathbf{Z}_t\mathbf{\Gamma}\boldsymbol{f}_{t+1}\right] \\ = \; & -2\text{tr}\left[\boldsymbol{f}_{t+1}^\top\mathbf{\Gamma}^\top\mathbf{Z}_t^\top\boldsymbol{r}_{t+1}\right] + \text{tr}\left[\mathbf{\Gamma}\boldsymbol{f}_{t+1}\boldsymbol{f}_{t+1}^\top\mathbf{\Gamma}^\top\mathbf{Z}_t^\top\mathbf{Z}_t\right] \\ = \; & -2\text{tr}\left[\mathbf{Z}_t^\top\boldsymbol{r}_{t+1}\boldsymbol{f}_{t+1}^\top\mathbf{\Gamma}^\top\right] + \text{tr}\left[\mathbf{\Gamma}\boldsymbol{f}_{t+1}\boldsymbol{f}_{t+1}^\top\mathbf{\Gamma}^\top\mathbf{Z}_t^\top\mathbf{Z}_t\right]. \end{aligned} \tag{79}$$

For the 1st term, let $\mathbf{A} = \mathbf{Z}_t^\top \boldsymbol{r}_{t+1}\boldsymbol{f}_{t+1}^\top$ and $\mathbf{B} = \boldsymbol{\Gamma}^\top$, and use the fact that $\text{tr}(\mathbf{AB}) = \text{vec}(\mathbf{A}^\top)^\top \text{vec}(\mathbf{B})$, then it becomes:

$$-2\text{tr}\left[\mathbf{Z}_t^\top \boldsymbol{r}_{t+1}\boldsymbol{f}_{t+1}^\top \boldsymbol{\Gamma}^\top\right] = -2\text{vec}(\boldsymbol{f}_{t+1}\boldsymbol{r}_{t+1}^\top \mathbf{Z}_t)^\top \text{vec}(\boldsymbol{\Gamma}^\top). \tag{80}$$

Now, let's further look at this part: $\text{vec}(\boldsymbol{f}_{t+1}\boldsymbol{r}_{t+1}^\top \mathbf{Z}_t)^\top$. Consider $\text{vec}(\boldsymbol{f}_{t+1}\boldsymbol{r}_{t+1}^\top \mathbf{Z}_t)$ without transpose ($\top$) first. Let $\mathbf{A} = \boldsymbol{f}_{t+1}$, $\mathbf{B} = \boldsymbol{r}_{t+1}^\top$, and $\mathbf{C} = \mathbf{Z}_t$, and use the fact that $\text{vec}(\mathbf{ABC}) = (\mathbf{C}^\top \otimes \mathbf{A})\text{vec}(\mathbf{B})$:

$$\text{vec}(\boldsymbol{f}_{t+1}\boldsymbol{r}_{t+1}^\top \mathbf{Z}_t) = (\mathbf{Z}_t^\top \otimes \boldsymbol{f}_{t+1})\text{vec}(\boldsymbol{r}_{t+1}^\top). \tag{81}$$

Note that $\text{vec}(\boldsymbol{r}_{t+1}^\top)$ transposes the vector $\boldsymbol{r}_{t+1}$ first and then vectorizes it (back), and therefore it is just $\boldsymbol{r}_{t+1}$. Hence:

$$\text{vec}(\boldsymbol{f}_{t+1}\boldsymbol{r}_{t+1}^\top \mathbf{Z}_t) = (\mathbf{Z}_t^\top \otimes \boldsymbol{f}_{t+1})\boldsymbol{r}_{t+1}. \tag{82}$$

Now, adding transpose to $\text{vec}(\boldsymbol{f}_{t+1}\boldsymbol{r}_{t+1}^\top \mathbf{Z}_t)$:

$$\begin{aligned} (\text{vec}(\boldsymbol{f}_{t+1}\boldsymbol{r}_{t+1}^\top \mathbf{Z}_t))^\top &= \boldsymbol{r}_{t+1}(\mathbf{Z}_t^\top \otimes \boldsymbol{f}_{t+1})^\top \\ &= \boldsymbol{r}_{t+1}(\mathbf{Z}_t \otimes \boldsymbol{f}_{t+1}^\top), \end{aligned} \tag{83}$$

where I use the fact that $(\mathbf{A} \otimes \mathbf{B})^\top = \mathbf{A}^\top \otimes \mathbf{B}^\top$. Therefore, for the 1st term, we eventually get:

$$-2\text{vec}(\boldsymbol{f}_{t+1}\boldsymbol{r}_{t+1}^\top \mathbf{Z}_t)^\top \text{vec}(\boldsymbol{\Gamma}^\top) = -2\boldsymbol{r}_{t+1}(\mathbf{Z}_t \otimes \boldsymbol{f}_{t+1}^\top)\text{vec}(\boldsymbol{\Gamma}^\top). \tag{84}$$

For the 2nd term, let $\mathbf{A}^\top = \boldsymbol{\Gamma}$, $\mathbf{B} = \boldsymbol{f}_{t+1}\boldsymbol{f}_{t+1}^\top$, $\mathbf{C} = \boldsymbol{\Gamma}^\top$, and $\mathbf{D}^\top = \mathbf{Z}_t^\top \mathbf{Z}_t$. Then, by $\text{tr}(\mathbf{A}^\top \mathbf{BCD}^\top) = \text{vec}(\mathbf{A})^\top (\mathbf{D} \otimes \mathbf{B})\text{vec}(\mathbf{C})$, the 2nd term becomes:

$$\text{vec}(\boldsymbol{\Gamma}^\top)^\top (\mathbf{Z}_t^\top \mathbf{Z}_t \otimes \boldsymbol{f}_{t+1}\boldsymbol{f}_{t+1}^\top)\text{vec}(\boldsymbol{\Gamma}^\top). \tag{85}$$

Combining both terms, the objective function becomes:

$$-2\boldsymbol{r}_{t+1}(\mathbf{Z}_t \otimes \boldsymbol{f}_{t+1}^\top)\text{vec}(\boldsymbol{\Gamma}^\top) + \text{vec}(\boldsymbol{\Gamma}^\top)^\top (\mathbf{Z}_t^\top \mathbf{Z}_t \otimes \boldsymbol{f}_{t+1}\boldsymbol{f}_{t+1}^\top)\text{vec}(\boldsymbol{\Gamma}^\top). \tag{86}$$

Stacking all $t = 1, \cdots, T-1$ periods and set the first order condition (F.O.C) with respect to $\text{vec}(\boldsymbol{\Gamma}^\top)$ to zero. Then:

$$\sum_{t=1}^{T-1} (\mathbf{Z}_t \otimes \boldsymbol{f}_{t+1}^\top)^\top \boldsymbol{r}_{t+1} = \left(\sum_{t=1}^{T-1} \mathbf{Z}_t^\top \mathbf{Z}_t \otimes \boldsymbol{f}_{t+1}\boldsymbol{f}_{t+1}^\top\right) \text{vec}(\boldsymbol{\Gamma}^\top). \tag{87}$$

Solving for $\text{vec}(\boldsymbol{\Gamma}^\top)$:

$$\text{vec}(\boldsymbol{\Gamma}^\top) = \left(\sum_{t=1}^{T-1} \mathbf{Z}_t^\top \mathbf{Z}_t \otimes \boldsymbol{f}_{t+1}\boldsymbol{f}_{t+1}^\top\right)^{-1} \left(\sum_{t=1}^{T-1} \left[\mathbf{Z}_t \otimes \boldsymbol{f}_{t+1}^\top\right]^\top \boldsymbol{r}_{t+1}\right). \tag{88}$$

These two conditions indicate that factor realizations are period-by-period cross section coef-

ficients of $\boldsymbol{r}_{t+1}$ on the latent loading matrix $\boldsymbol{\Gamma}$; while $\boldsymbol{\Gamma}$ is the coefficient of returns regressed on the factors interacted with firm-specific characteristics. Kelly, Pruitt, and Su (2019) mention that this system of first-order conditions has no closed-form solution and therefore it must be solved numerically, and it is solvable via alternating least squares efficiently. (Hopefully, now you see why we dive into ALS in great detail in Section 3!)

6.4.3 Intuitive Explanation

In his talk at the 2018 Utah Winter Finance Conference, Bryan Kelly offered a managed portfolio interpretation for the IPCA model, which I find very intuitive. The IPCA estimator is approximately equivalent to:

- For each characteristic $l = 1, \cdots, L$ in each period t:
  - ⋆ Run a cross-section regression of $R_{i,t}$ on $Z_{i,l,t-1}$, estimates $X_{l,t}$,
  - ⋆ $X_{l,t}$ is return of the managed portfolio associated with characteristic l.
- From $T \times L$ matrix $\mathbf{X}$ of characteristic-managed portfolio returns:
  - ⋆ IPCA factor estimates $\approx$ first K PCs.

According to this interpretation, IPCA factors reflect common sources of variation in managed portfolios, and the estimated factors are "portfolios of managed portfolios".

7 A New Norm Beyond PCA

The most important feature of IPCA is that it brings firm characteristics into the latent factor model. Yet, it is still restrictive to some extent because it only considers linear mappings from firm characteristics to factor exposures. What if we consider nonlinear mappings? That brings us to a new research norm for empirical asset pricing.

Consider again the (conditional) latent factor model:

$$R_{i,t+1} = \boldsymbol{\beta}(\mathbf{C})_{i,t}^{\top} \boldsymbol{f}_{t+1} + e_{i,t+1}, \tag{89}$$

where $\mathbf{C}$ represent firm characteristics that are related to asset returns. In this framework, the work goes into modeling factor exposure $\boldsymbol{\beta}$. From a **machine learning** perspective, $\boldsymbol{\beta}()$ maps a large information set into a small factor model ($\mathbb{R}^L \to \mathbb{R}^K, L \gg K$). Once we find $\boldsymbol{\beta}()$, the optimal factor estimates can be found by solving the least squares, i.e., $\boldsymbol{f}_{t+1} = (\boldsymbol{\beta}_t^{\top}\boldsymbol{\beta}_t)^{-1}\boldsymbol{\beta}_t^{\top}\boldsymbol{R}_{t+1}$.

Within this framework, the IPCA model can be viewed as linearly mapping firm characteristics to $\boldsymbol{\beta}$ via $\boldsymbol{\Gamma}$. However, it is evident that we need not limit ourselves to linear models. In terms of nonlinear models, the autoencoder neural network model proposed by Gu, Kelly, and Xiu (2021) serves as a representative example. Furthermore, through asset pricing theory, we can demonstrate

that the factor model approach is equivalent to the Stochastic Discount Factor (SDF) framework. Therefore, within a unified framework (either from the SDF view or from the factor model view), we can analyze and compare the various empirical asset pricing models based on machine learning algorithms proposed in recent years by academia.

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